



Time Series Analysis and Forecasting

Chapter 5: GARCH and Volatility



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Learning Objectives

By the end of this chapter, you will be able to:

1. Understand **volatility clustering** and stylized facts of financial returns
2. Estimate and interpret **ARCH** and **GARCH** models
3. Apply asymmetric models (**EGARCH**, **GJR-GARCH**) for the leverage effect
4. Perform model validation and selection
5. Forecast volatility and calculate **Value at Risk (VaR)**

Practical Skills

- Python implementation with the `arch` package
 - ▶ Estimation, forecasting, and automatic diagnostics
- Interpreting parameters and volatility persistence
- VaR calculation for risk management
 - ▶ Backtesting and forecast validation

Outline

Foundations

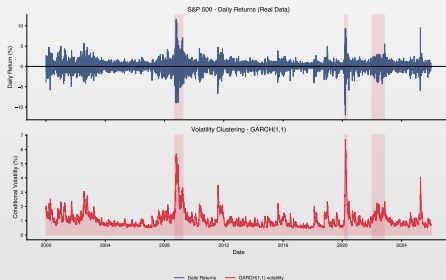
- ▣ Motivation
- ▣ Introduction to Volatility Modeling
- ▣ The ARCH Model
- ▣ The GARCH Model
- ▣ Asymmetric GARCH Models
- ▣ Model Selection and Diagnostics

Applications

- ▣ Volatility Forecasting
- ▣ Python Implementation
- ▣ Case Study: S&P 500
- ▣ Case Study: Bitcoin
- ▣ Summary and Quiz

Volatility Clustering

- High volatility periods are followed by high volatility; calm periods by calm periods
- **Takeaway:** Volatility clustering is visible **even before modeling** — it is the first sign that variance is not constant



Why Model Volatility?

Empirical Observations in Financial Series

- ▣ Financial returns exhibit **volatility clustering** — periods of high volatility tend to be followed by periods of high volatility
- ▣ The distribution of returns has **fat tails** (leptokurtosis)
- ▣ Return correlation is nearly zero, but correlation of squares is significant
- ▣ Volatility responds **asymmetrically** to shocks (leverage effect)

Central Idea of this Chapter

- ▣ **Even though returns are hard to predict, their volatility is often predictable.**
- ▣ ARIMA models assume **constant variance** (homoskedasticity) — an unrealistic assumption for financial series

Why Do We Care About Conditional Volatility?

Direct Applications in Finance

- ▣ **Value at Risk (VaR) and Expected Shortfall (ES):** Basel III/IV capital requirements
- ▣ **Option pricing:** implied vs realized volatility
- ▣ **Dynamic hedging:** optimal hedge ratios $h_t^* = \sigma_{12,t}/\sigma_{22,t}$
- ▣ **Margining:** clearing houses recalculate margins daily
- ▣ **Economic/regulatory capital:** FRTB, stress testing
- ▣ **Portfolio allocation:** Markowitz weights with time-varying H_t

Conclusion

- ▣ Conditional volatility is not an academic exercise — it is the **central ingredient** of quantitative risk management

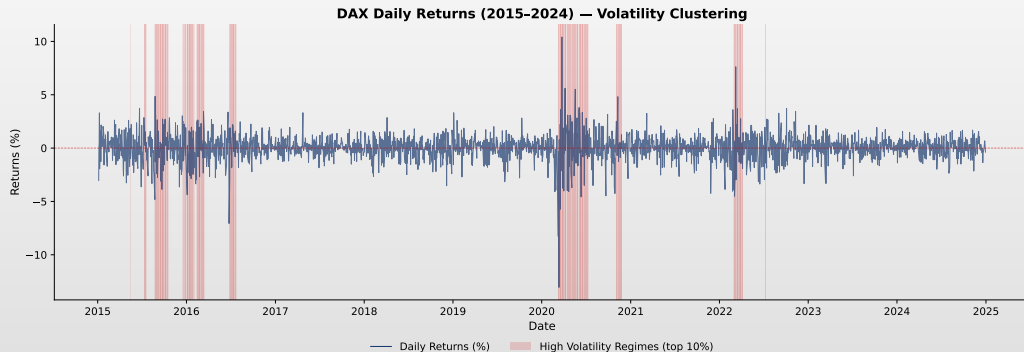
Example: DAX – Volatility Clustering

Observations

- ▣ DAX daily returns (2015–2024): pronounced volatility clustering
 - ▶ Extreme volatility episodes during crises (Brexit 2016, COVID 2020, Ukraine war 2022)
- ▣ Red zones mark high-volatility regimes (top 10% of 21-day rolling volatility)
 - ▶ Calm and turbulent periods alternate — they do not appear randomly

Example: DAX – Volatility Clustering

- **Takeaway:** Daily returns show calm/crisis alternation — a constant-variance model cannot capture this phenomenon

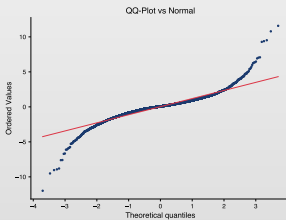
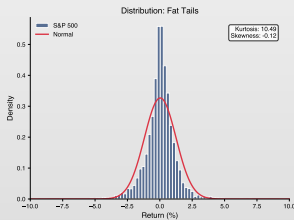
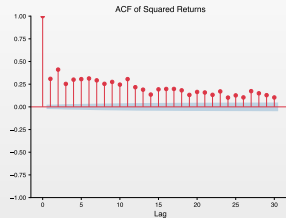
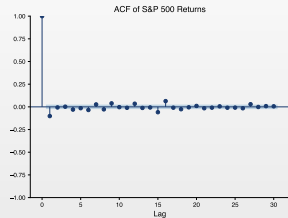


Stylized Facts of Financial Returns

Observed Properties

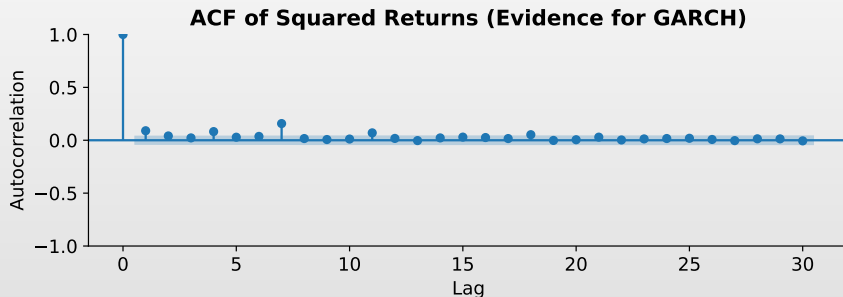
- ▣ **No autocorrelation** in returns
 - ▶ $\text{Corr}(r_t, r_{t-k}) \approx 0$ — returns are unpredictable
- ▣ **Autocorrelation** in $r_t^2, |r_t|$
 - ▶ $\text{Corr}(r_t^2, r_{t-k}^2) > 0$ — volatility is predictable
- ▣ **Fat tails** (kurtosis > 3)
 - ▶ More extreme values than the normal distribution predicts
- ▣ **Leverage effect**
 - ▶ Negative shocks increase volatility more than positive shocks
- ▣ **Volatility clustering**
 - ▶ As discussed above, conditional variance is predictable and persistent

Stylized Facts of Financial Returns



Example: Bitcoin – Evidence for ARCH Effects

- ▣ $ACF(r_t^2)$ significant at many lags $\Rightarrow$ **ARCH effects present**
- ▣ **Takeaway:** ACF on squares suggests dependence in the second moment — even when ACF on returns ≈ 0



Conditional Heteroskedasticity

Definition 1 (Conditional Variance)

For return series $\{r_t\}$, the **conditional variance** at time t is: $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1}) = \mathbb{E}[(r_t - \mu_t)^2 | \mathcal{F}_{t-1}]$ where $\mathcal{F}_{t-1}$ is the information available up to time $t - 1$.

General Model

$$r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d.}(0, 1)$$

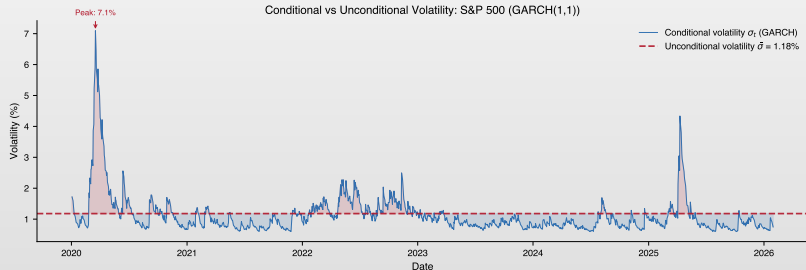
- μ_t = conditional mean (ARMA); σ_t^2 = conditional variance (GARCH)
- z_t = standardized innovations (Normal, Student-t, GED)



Conditional vs Unconditional Volatility

Interpretation

- Conditional volatility σ_t (GARCH): time-varying, reacts to new information
- Unconditional volatility $\bar{\sigma} = \sqrt{\omega/(1 - \alpha - \beta)}$: constant long-run level
- Red zones: $\sigma_t > \bar{\sigma}$ (stress periods); blue zones: $\sigma_t < \bar{\sigma}$ (calm periods)



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Introduction to Volatility Modeling

The ARCH Model

The GARCH Model

Asymmetric GARCH Models

Model Selection and Diagnostics

Volatility Forecasting

Python Implementation

Case Study: S&P 500

Case Study: Bitcoin

AI Use Case

Summary

Quiz

Researcher Spotlight: Engle & Bollerslev



Robert Engle (*1942)

Nobel Prize 2003



Tim Bollerslev (*1958)



Biography

- ▣ **Robert Engle:** American economist at NYU Stern. Nobel Prize (2003) “for methods of analyzing economic time series with time-varying volatility (ARCH)”
- ▣ **Tim Bollerslev:** Danish-American economist at Duke University, PhD student of Engle

Key Contributions

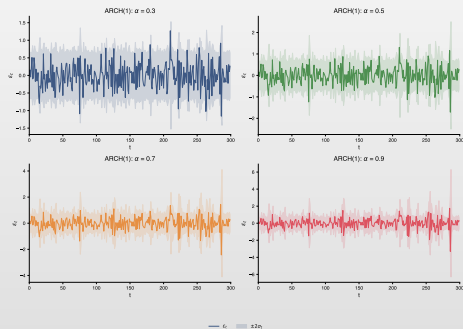
- ▣ **ARCH model** (Engle, 1982) — autoregressive conditional heteroskedasticity
- ▣ **GARCH model** (Bollerslev, 1986) — generalized ARCH with persistent volatility
- ▣ **Realized volatility** measures and high-frequency econometrics
- ▣ Foundation for modern financial risk management (VaR, ES)

This is the foundation of modern volatility econometrics and quantitative risk management.

ARCH(1) Simulation: Effect of α Parameter

Interpretation

- Higher α means volatility reacts more strongly to recent shocks



ARCH(1) Simulation: What Do We Observe?

Reading the Simulation Plot

- ▣ **Small** α (e.g., 0.1): series looks almost like white noise — mild clustering, rare spikes
- ▣ **Moderate** α (e.g., 0.5): visible spikes, but they die out quickly (1–2 periods) — “short memory”
- ▣ **Large** α (e.g., 0.9): explosive volatility — very large spikes, but still no real persistence

Key Difference: ARCH vs GARCH

- ▣ ARCH reacts only to the **last shock** (ε_{t-1}^2) — volatility rises sharply but falls just as sharply. It cannot capture the long high-volatility episodes observed in markets. GARCH adds $\beta\sigma_{t-1}^2 =$ “memory”.

Visual Example

- ▣ Compare: in the ARCH(1) simulation, the $\pm 2\sigma$ bands have narrow spikes (1 day). On real S&P 500, volatility episodes last weeks $\Rightarrow$ ARCH is insufficient.

The ARCH(q) Model — Engle (1982)

Definition 2 (ARCH(q))

The **Autoregressive Conditional Heteroskedasticity** model of order q :

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d.}(0, 1), \quad \sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$$

Stationarity Restrictions

- ▣ $\omega > 0$ (positive base variance), $\alpha_i \geq 0$ (non-negativity)
- ▣ $\sum_{i=1}^q \alpha_i < 1$ (stationarity)

Properties of the ARCH(1) Model

$$\text{ARCH}(1): \sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2$$

- **Unconditional variance:** $\mathbb{E}[\varepsilon_t^2] = \frac{\omega}{1 - \alpha_1}$ (if $\alpha_1 < 1$)
- **Kurtosis:** $\kappa = 3 \cdot \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2}$ (if $\alpha_1^2 < 1/3$)
- Kurtosis > 3 for $\alpha_1 > 0 \Rightarrow$ **fat tails!**

Numerical Example

If $\omega = 0.0001$ and $\alpha_1 = 0.3$:

- Unconditional variance: $\sigma^2 = \frac{0.0001}{1 - 0.3} = 0.000143$
- Kurtosis: $\kappa = 3 \cdot \frac{1 - 0.09}{1 - 0.27} = 3.74 > 3$

Derivation: Unconditional Variance of ARCH(1)

Derivation.

Let $\varepsilon_t = \sigma_t z_t$ with $z_t \sim N(0, 1)$ and $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2$.

Step 1: Take unconditional expectation:

$$\mathbb{E}[\varepsilon_t^2] = \mathbb{E}[\sigma_t^2 z_t^2] = \mathbb{E}[\sigma_t^2] \cdot \mathbb{E}[z_t^2] = \mathbb{E}[\sigma_t^2] \quad (\text{independence: } \sigma_t^2 \text{ is } \mathcal{F}_{t-1}\text{-measurable, } z_t \perp \mathcal{F}_{t-1})$$

Step 2: Apply expectation to variance equation:

$$\mathbb{E}[\sigma_t^2] = \mathbb{E}[\omega + \alpha_1 \varepsilon_{t-1}^2] = \omega + \alpha_1 \mathbb{E}[\varepsilon_{t-1}^2]$$

Step 3: By stationarity, $\mathbb{E}[\varepsilon_t^2] = \mathbb{E}[\varepsilon_{t-1}^2] = \sigma^2$:

$$\sigma^2 = \omega + \alpha_1 \sigma^2 \quad \Rightarrow \quad \sigma^2(1 - \alpha_1) = \omega$$

Result: $\boxed{\sigma^2 = \frac{\omega}{1 - \alpha_1}}$ (requires $\alpha_1 < 1$ for stationarity)



Derivation: Kurtosis of ARCH(1)

For $\varepsilon_t = \sigma_t z_t$ with $z_t \sim N(0, 1)$:

Step 1: $\mathbb{E}[\varepsilon_t^4] = \mathbb{E}[\sigma_t^4] \cdot \mathbb{E}[z_t^4] = 3\mathbb{E}[\sigma_t^4]$ (since $\mathbb{E}[z^4] = 3$)

Step 2: Using $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2$:

$$\mathbb{E}[\sigma_t^4] = \mathbb{E}[(\omega + \alpha_1 \varepsilon_{t-1}^2)^2] = \omega^2 + 2\omega\alpha_1\sigma^2 + \alpha_1^2\mathbb{E}[\varepsilon_{t-1}^4]$$

Step 3: Solving the recursion yields:

$$\kappa = \frac{\mathbb{E}[\varepsilon_t^4]}{(\mathbb{E}[\varepsilon_t^2])^2} = 3 \cdot \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2}$$

Interpretation

- ▣ $\kappa > 3$ for any $\alpha_1 > 0 \Rightarrow$ **fat tails** (leptokurtosis)
- ▣ Requires $\alpha_1 < 0.577$ for finite fourth moment
- ▣ ARCH naturally generates heavy-tailed distributions!

Derivation: Kurtosis of GARCH(1,1)

For $\varepsilon_t = \sigma_t z_t$ with $z_t \sim N(0, 1)$ and $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$:

Step 1: $\mathbb{E}[\varepsilon_t^4] = \mathbb{E}[\sigma_t^4 z_t^4] = 3\mathbb{E}[\sigma_t^4]$ (since $\mathbb{E}[z^4] = 3$)

Step 2: Using $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, square both sides:

$$\mathbb{E}[\sigma_t^4] = \mathbb{E}[(\omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2)^2]$$

Expand and use stationarity ($\mathbb{E}[\sigma_t^4] = \mathbb{E}[\sigma_{t-1}^4]$, $\mathbb{E}[\varepsilon_t^4] = 3\mathbb{E}[\sigma_t^4]$):

$$\mathbb{E}[\sigma_t^4] = \omega^2 + 2\omega(\alpha + \beta)\bar{\sigma}^2 + (3\alpha^2 + 2\alpha\beta + \beta^2)\mathbb{E}[\sigma_{t-1}^4]$$

Step 3: Solving the recursion:

$$\kappa_\varepsilon = \frac{\mathbb{E}[\varepsilon_t^4]}{(\mathbb{E}[\varepsilon_t^2])^2} = 3 \cdot \frac{1 - (\alpha + \beta)^2}{1 - (\alpha + \beta)^2 - 2\alpha^2}$$

Interpretation

- ▣ Requires $1 - (\alpha + \beta)^2 - 2\alpha^2 > 0$ for finite fourth moment
- ▣ GARCH generates **more kurtosis** than ARCH because $\beta > 0$ amplifies persistence
- ▣ With $\alpha = 0.088$, $\beta = 0.900$: $\kappa \approx 5.6$ (vs. ARCH(1): $\kappa \approx 3.05$)

Theory vs Empirics: The Kurtosis Gap

ARCH(1) Kurtosis: Too Low!

- For typical stock returns: $\alpha_1 \approx 0.09$ (S&P 500)
- Theoretical kurtosis: $\kappa = 3 \cdot \frac{1-0.09^2}{1-3 \times 0.09^2} \approx \mathbf{3.05}$
- Empirical kurtosis of S&P 500 daily returns: $\kappa \approx \mathbf{13.8}$

The Kurtosis Gap

$$\underbrace{3.05}_{\text{ARCH(1)-Normal}} \ll \underbrace{13.8}_{\text{Empirical}}$$

ARCH(1) with Gaussian innovations cannot match the extreme kurtosis observed in financial data.

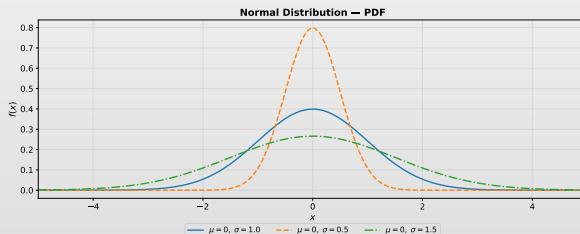
Solutions

- **Student- t innovations:** $z_t \sim t_\nu$ adds kurtosis from the distribution itself
- **GARCH(1,1):** persistence ($\beta > 0$) generates additional kurtosis
- Both combined: Student- t GARCH captures $\kappa \gg 3$ observed in practice

The Normal (Gaussian) Distribution

Probability Density Function (PDF)

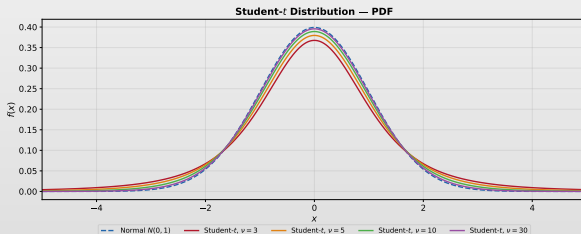
- $f(x | \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
- Parameters: μ (mean), $\sigma > 0$ (standard deviation)
- Kurtosis: $\kappa = 3$ (reference — the mesokurtic distribution)
- Limitation: **underestimates** the frequency of extreme values in financial series



The Student- t Distribution

Probability Density Function (PDF)

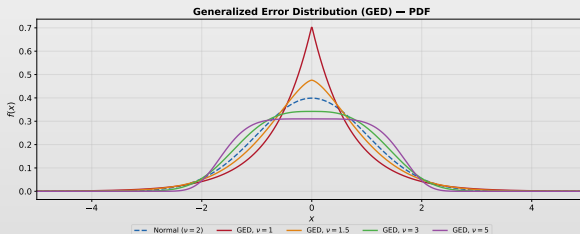
- $f(x | \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$
- Parameter: $\nu > 2$ (degrees of freedom); finite variance only if $\nu > 2$
- Kurtosis: $\kappa = 3 + \frac{6}{\nu-4}$ (for $\nu > 4$); $\nu \rightarrow \infty \Rightarrow N(0, 1)$
- **Advantage:** fat tails controlled by ν — captures extreme values in financial series



The GED (Generalized Error Distribution)

Probability Density Function (PDF)

- $f(x | \nu) = \frac{\nu \exp\left(-\frac{1}{2} \left|\frac{x}{\lambda}\right|^\nu\right)}{\lambda \cdot 2^{1+1/\nu} \cdot \Gamma(1/\nu)}$, $\lambda = \sqrt{2^{-2/\nu} \cdot \Gamma(1/\nu) / \Gamma(3/\nu)}$
- Parameter: $\nu > 0$ (shape); $\nu = 2$: Normal; $\nu = 1$: Laplace; $\nu \rightarrow \infty$: Uniform
- Kurtosis: $\kappa = \frac{\Gamma(1/\nu) \Gamma(5/\nu)}{\Gamma(3/\nu)^2}$; $\nu < 2$: fat tails; $\nu > 2$: thin tails
- **Advantage:** maximum flexibility — a single parameter controls the entire distribution shape

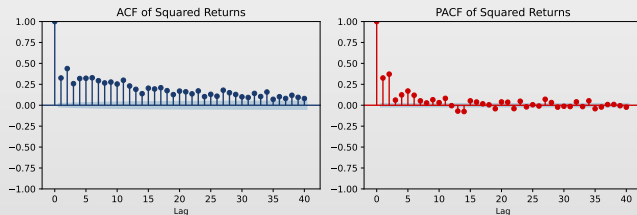


Correlogram of Squared Residuals

Visual Diagnostic for ARCH Effects

- If $\hat{\varepsilon}_t$ are i.i.d., then $\hat{\varepsilon}_t^2$ should show no autocorrelation
- Significant ACF of $\hat{\varepsilon}_t^2$ at multiple lags $\Rightarrow$ **conditional heteroskedasticity**
- Complements the Engle LM test with an intuitive visualization

S&P 500 — Autocorrelation of Squared Returns



Testing for ARCH Effects

Engle's Test for ARCH Effects

Procedure:

1. Estimate the mean model and obtain residuals $\hat{\varepsilon}_t$
2. Calculate $\hat{\varepsilon}_t^2$
3. Regress $\hat{\varepsilon}_t^2$ on its lags:

$$\hat{\varepsilon}_t^2 = \beta_0 + \beta_1 \hat{\varepsilon}_{t-1}^2 + \cdots + \beta_q \hat{\varepsilon}_{t-q}^2 + u_t$$

4. Calculate the statistic $LM = T \cdot R^2 \sim \chi^2(q)$

Hypotheses

- H_0 : No ARCH effects ($\alpha_1 = \cdots = \alpha_q = 0$)
- H_1 : ARCH effects present (at least one $\alpha_i \neq 0$)

Limitations of the ARCH Model

Practical Problems

1. **High order** — many lags are usually needed (large q)
2. **Many parameters** — estimation difficulties
3. **Non-negativity constraints** — difficult to impose for large q
4. **Does not capture persistence** — observed volatility is very persistent

The Solution

The GARCH Model — introduces lags of conditional variance to capture persistence with fewer parameters!

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Case Study: Bitcoin

AI Use Case

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The GARCH(1,1) Model

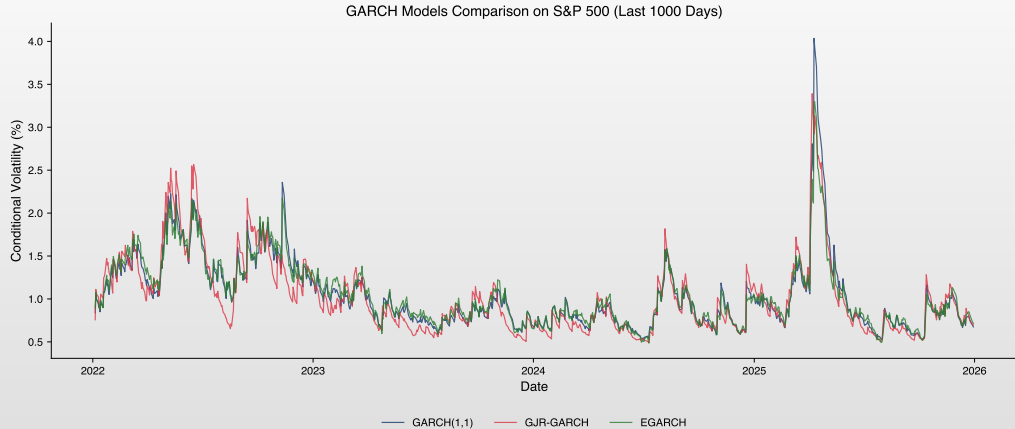
The Most Popular Volatility Model

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Restrictions & Properties

- ▣ $\omega > 0, \alpha \geq 0, \beta \geq 0; \quad \alpha + \beta < 1$ (stationarity)
- ▣ $\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}; \quad \text{Half-life: } HL = \frac{\ln(0.5)}{\ln(\alpha+\beta)}$

The GARCH(1,1) Model



The GARCH(p,q) Model — Bollerslev (1986)

Definition 3 (GARCH(p,q))

The **Generalized ARCH** model:

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d.}(0, 1)$$
$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Interpretation

- ▣ ω = base level of volatility
- ▣ α_i = reaction to recent shocks (news coefficients)
- ▣ β_j = volatility persistence (memory)
- ▣ $\alpha + \beta$ = total persistence

How to Read a GARCH Model in 30 Seconds

Structure of a GARCH(1,1) Model

- **Mean equation:** $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, $z_t \sim D(0, 1)$
- **Variance equation:** $\sigma_t^2 = \underbrace{\omega}_{\text{base}} + \underbrace{\alpha \varepsilon_{t-1}^2}_{\text{shock reaction}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{persistence}}$

Economic Interpretation of Parameters

- Large $\alpha \Rightarrow$ market **very sensitive** to new information (sudden shocks)
- Large $\beta \Rightarrow$ volatility shocks **dissipate slowly** (inertia)
- $\alpha + \beta$ close to 1 $\Rightarrow$ **very persistent** volatility (near-IGARCH)
- $\omega / (1 - \alpha - \beta) =$ unconditional variance (long-run equilibrium level)

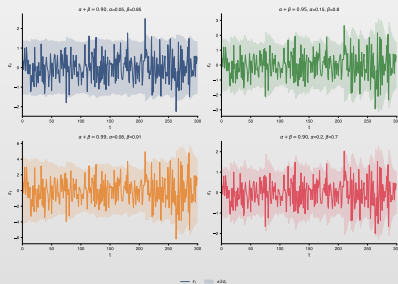
Choosing the Innovation Distribution z_t

- **Normal:** simple, but underestimates tail risk
- **Student-t:** fatter tails $\Rightarrow$ more realistic VaR; **standard choice** in practice

GARCH(1,1) Simulation: Persistence Effect

Interpretation

- α controls reaction to shocks
- β controls persistence
- The sum $\alpha + \beta$ determines mean-reversion speed



GARCH(1,1) Simulation: What Do We Observe?

Reading the Simulation Plot

- ▣ **Small** $\alpha + \beta$ (0.7): volatility reverts quickly — series looks almost like white noise
- ▣ **Moderate** $\alpha + \beta$ (0.95): high-volatility episodes last weeks — similar to real financial series
- ▣ $\alpha + \beta \approx 1$ (0.99): volatility persists for months — half-life ≈ 69 days

Visual Identification: Half-Life

- ▣ Half-life: $HL = \ln(0.5) / \ln(\alpha + \beta)$
- ▣ $\alpha + \beta = 0.95$: $HL \approx 14$ days (2 wks) $\alpha + \beta = 0.988$: $HL \approx 58$ days (3 mo, typical S&P 500)

Practical Rule

If the $\pm 2\sigma_t$ bands do not revert to normal within 1–2 months, the market has high persistence ($\alpha + \beta > 0.97$)

Derivation: Unconditional Variance of GARCH(1,1)

Derivation.

For $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$:

Step 1: Take unconditional expectation: $\mathbb{E}[\sigma_t^2] = \omega + \alpha \mathbb{E}[\varepsilon_{t-1}^2] + \beta \mathbb{E}[\sigma_{t-1}^2]$

Step 2: By stationarity, $\mathbb{E}[\sigma_t^2] = \mathbb{E}[\sigma_{t-1}^2] = \bar{\sigma}^2$ and $\mathbb{E}[\varepsilon_t^2] = \bar{\sigma}^2$: $\bar{\sigma}^2 = \omega + (\alpha + \beta)\bar{\sigma}^2$

Step 3: Solve: $\bar{\sigma}^2(1 - \alpha - \beta) = \omega \Rightarrow \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$



Stationarity Condition

Requires $\alpha + \beta < 1$ for finite unconditional variance.



GARCH(1,1) as ARMA for ε_t^2

ARMA(1,1) Representation

Define $\nu_t = \varepsilon_t^2 - \sigma_t^2$ (variance shock). Then:

$$\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + \nu_t - \beta\nu_{t-1}$$

This is an **ARMA(1,1)** for ε_t^2 !

Implications

- ACF of ε_t^2 decays exponentially (like ARMA)
- Persistence is given by $\alpha + \beta$
- PACF can help identify the order

Derivation: ARMA Representation of GARCH(1,1)

Derivation.

Step 1: Define variance shock: $\nu_t = \varepsilon_t^2 - \sigma_t^2$

□ $\mathbb{E}[\nu_t | \mathcal{F}_{t-1}] = \mathbb{E}[\varepsilon_t^2 | \mathcal{F}_{t-1}] - \sigma_t^2 = \sigma_t^2 - \sigma_t^2 = 0$

□ ν_t is a martingale difference sequence

Step 2: Substitute $\sigma_t^2 = \varepsilon_t^2 - \nu_t$ into GARCH equation:

$$\varepsilon_t^2 - \nu_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta (\varepsilon_{t-1}^2 - \nu_{t-1})$$

Step 3: Rearrange:

$$\varepsilon_t^2 = \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + \nu_t - \beta \nu_{t-1}$$

Result: ARMA(1,1) with AR coefficient $\phi = \alpha + \beta$ and MA coefficient $\theta = -\beta$. □

Derivation: Volatility Persistence and Half-Life

Multi-Step Forecast GARCH(1,1)

$$\square \mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$$

Derivation

- **Step 1:** Let $\phi = \alpha + \beta$ and $q_t = \sigma_t^2 - \bar{\sigma}^2$ (deviation from mean)
- **Step 2:** From the GARCH equation: $\mathbb{E}_t[q_{t+1}] = \phi \cdot q_t$, so $\mathbb{E}_t[q_{t+h}] = \phi^h \cdot q_t$
- **Step 3:** Half-life = time until deviation halves: $\phi^{HL} = 0.5 \Rightarrow HL = \frac{\ln(0.5)}{\ln(\phi)} = \frac{-0.693}{\ln(\alpha+\beta)}$

Example: S&P 500

- With $\alpha + \beta = 0.988$: $HL = \frac{-0.693}{-0.012} \approx \mathbf{58 \text{ days}}$ (shocks persist ~ 3 months!)

Why GARCH works: the core intuition

The fundamental idea

Financial returns are hard to predict, but their volatility is predictable:

$$r_t = \sigma_t \cdot z_t, \quad z_t \sim \text{i.i.d.}(0, 1)$$

- r_t is unpredictable (efficient market), but σ_t^2 evolves slowly
- GARCH captures this: **shocks affect volatility for many periods**

The GARCH(1,1) mechanism

$$\sigma_t^2 = \underbrace{\omega}_{\text{equilibrium level}} + \underbrace{\alpha r_{t-1}^2}_{\text{new information (shock)}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{volatility memory}}$$

- Was yesterday turbulent? (r_{t-1}^2 large) $\Rightarrow$ today's volatility rises
- Was volatility already high? (σ_{t-1}^2 large) $\Rightarrow$ it persists today
- Long-run: volatility reverts to equilibrium $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$

Why it matters

- Without GARCH: constant VaR $\Rightarrow$ **underestimates** risk in crises, **overestimates** in calm periods
- With GARCH: dynamic VaR $\Rightarrow$ **adapts** to market conditions $\Rightarrow$ realistic risk management

Estimation of GARCH Models

Maximum Likelihood Estimation (MLE)

$$\text{Log-likelihood (normal): } \ell(\theta) = -\frac{T}{2} \ln(2\pi) - \frac{1}{2} \sum_{t=1}^T \left[\ln(\sigma_t^2) + \frac{\varepsilon_t^2}{\sigma_t^2} \right]$$

Alternative Distributions for z_t

- ▣ **Student-t**: captures fat tails — most common choice
- ▣ **GED**: flexibility for kurtosis
- ▣ **Skewed Student-t**: asymmetry and fat tails

Practical Note

Student-t distribution typically provides better fit for financial returns due to fat tails (kurtosis > 3).

Student- t Log-Likelihood

Why Student- t ?

- ▣ Gaussian innovations underestimate fat tails: kurtosis fixed at 3
- ▣ Student- t adds a parameter $\nu > 2$ (degrees of freedom) that controls tail thickness

Definition 4 (Standardized Student- t Log-Likelihood)

$$\ell_t(\theta, \nu) = \ln \Gamma\left(\frac{\nu+1}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln[(\nu-2)\pi\sigma_t^2] - \frac{\nu+1}{2} \ln\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)$$

where $\Gamma(\cdot)$ is the Gamma function, $\nu > 2$, and σ_t^2 is from the GARCH equation.

Properties

- ▣ When $\nu \rightarrow \infty$: converges to the Normal log-likelihood
- ▣ Student- t kurtosis: $3(\nu-2)/(\nu-4)$ for $\nu > 4$ (e.g., $\nu = 6 \Rightarrow \kappa = 6$)
- ▣ ν is estimated jointly with (ω, α, β) via MLE

Choosing the Innovation Distribution

Normal

- Simple, fast
- Kurtosis = 3 (fixed)
- Underestimates extremes
- VaR 1%: 2.326σ

Student- t

- One extra parameter (ν)
- Adjustable kurtosis
- Realistic fat tails
- VaR 1%: $> 2.326\sigma$

GED

- Shape parameter (ν)
- Includes Normal ($\nu = 2$)
- Flexible, but less intuitive
- Rarely used in practice

Conclusion

- In risk management, the choice of distribution is almost as important as the choice of variance dynamics. Student- t is the standard choice.

Quasi-MLE and Robustness

What if the Distribution is Misspecified?

- ▣ Standard MLE assumes $z_t \sim N(0, 1)$, but true distribution may differ
- ▣ **Quasi-MLE (QMLE)**: maximize the Gaussian log-likelihood even if z_t is non-normal
- ▣ Key insight: we only need the *first two conditional moments* to be correct

Definition 5 (QMLE Consistency — Weiss (1986); Lee & Hansen (1994))

If the conditional mean and variance are correctly specified, QMLE is consistent for (ω, α, β) regardless of the true distribution of z_t . Robust (sandwich) standard errors: Bollerslev & Wooldridge (1992).

Sandwich (Robust) Standard Errors

Standard errors must be corrected: $\hat{V}_{\text{robust}} = \hat{A}^{-1} \hat{B} \hat{A}^{-1}$ (White, 1982), where $\hat{A}$ = Hessian, $\hat{B}$ = outer product of scores. In Python: `result.summary()` reports robust SEs by default.

When to Use QMLE vs Full MLE

- ▣ **QMLE**: robust default — consistent even under misspecification
- ▣ **Full MLE** (Student- t): more *efficient* if the distribution is correct

Practical Estimation Challenges

Common Issues

- ▣ **Convergence failure:** Log-likelihood is non-convex; optimizer may get stuck
- ▣ **Boundary solutions:** $\hat{\alpha} \approx 0$ or $\hat{\alpha} + \hat{\beta} \approx 1$ (near-IGARCH)
- ▣ **Sensitivity to starting values:** Different initializations $\Rightarrow$ different estimates

Best Practices

1. Try multiple optimizers: **BFGS** (fast, gradient-based) vs **Nelder-Mead** (derivative-free)
2. Initialize σ_0^2 with sample variance (backcast)
3. Check $\alpha + \beta < 1$ post-estimation; if ≈ 1 , consider IGARCH or regime-switching
4. Compare results across starting values; consistent estimates $\Rightarrow$ reliable model

Rule of Thumb

If BFGS and Nelder-Mead give *different* parameter estimates, the model may be misspecified or the data insufficient.

Typical Values for GARCH(1,1)

Series	α	β	$\alpha + \beta$
S&P 500 daily	0.05–0.10	0.85–0.95	0.95–0.99
EUR/USD daily	0.03–0.08	0.90–0.95	0.95–0.99
Bitcoin daily	0.10–0.20	0.75–0.85	0.90–0.98
Bonds	0.02–0.05	0.90–0.97	0.95–0.99

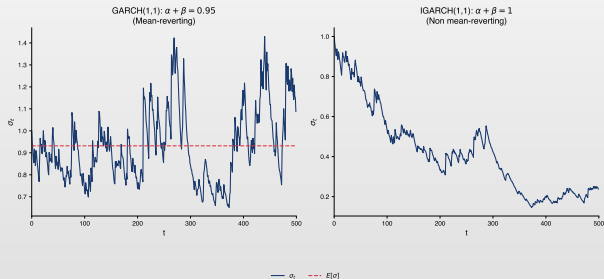
Observations

- ▣ $\alpha + \beta$ close to 1 $\Rightarrow$ **very persistent volatility**
- ▣ Small α , large $\beta \Rightarrow$ slow reaction to shocks, long memory
- ▣ Bitcoin: larger $\alpha \Rightarrow$ faster reaction to news

GARCH vs IGARCH: Persistence Comparison

Interpretation

- Standard GARCH reverts to unconditional mean
- IGARCH has no finite mean $\Rightarrow$ shocks persist indefinitely



IGARCH: What Do We Observe in the Simulated Series?

Reading the Comparison Plot

- ▣ **Standard GARCH** ($\alpha + \beta < 1$): volatility reverts to $\bar{\sigma}^2$ — “waves” with decreasing amplitude
- ▣ **IGARCH** ($\alpha + \beta = 1$): no anchor — after a large shock, the elevated level persists indefinitely
- ▣ Visually: IGARCH “wanders” (random walk in variance), while GARCH oscillates around the mean

How to Detect IGARCH in Practice?

- ▣ Estimate GARCH(1,1) and check: if $\hat{\alpha} + \hat{\beta} > 0.99$, consider IGARCH
- ▣ ACF of ε_t^2 decays very slowly (like a long-memory process)
- ▣ Frequently encountered in currency markets (FX) and some commodities

Caution

IGARCH may be an artifact of regime changes — verify with Markov-switching models!

IGARCH — Integrated GARCH

TSA_ch5_igarch

Definition 6 (IGARCH(1,1))

When $\alpha + \beta = 1$:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + (1 - \alpha) \sigma_{t-1}^2$$

Properties

- Unconditional variance does not exist (infinite) (but the process remains strictly stationary, cf. Nelson, 1990)
- Shocks have **permanent** effect on volatility
- Used for series with extreme persistence
- Useful for **RiskMetrics** (J.P. Morgan): $\alpha = 0.06$, $\beta = 0.94$
- IGARCH is analogous to a **unit root in variance**

Nelson's Theorem (1990): Strict Stationarity of IGARCH

TSA_ch5_nelson_theorem

Theorem 1 (Nelson, 1990)

Let the IGARCH(1,1) process: $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + (1 - \alpha) \sigma_{t-1}^2$, z_t i.i.d., $\mathbb{E}[z_t^2] = 1$. If $\omega > 0$ and $\mathbb{E}[\ln(\alpha z_t^2 + (1 - \alpha))] < 0$, then the process $\{\varepsilon_t\}$ is strictly stationary and ergodic, even though the unconditional variance is infinite.

Lyapunov Condition

- ▣ The condition $\mathbb{E}[\ln(\alpha z_t^2 + (1 - \alpha))] < 0$ is equivalent to a **negative top Lyapunov exponent**
- ▣ Verified for standard distributions ($z_t \sim \mathcal{N}(0, 1)$, Student- t) when $\alpha \in (0, 1)$
- ▣ Interpretation: log-volatility $\ln \sigma_t^2$ has a mean-contracting tendency

Proof Sketch

- ▣ Rewrite σ_t^2 as a stochastic recurrence: $Y_t = A_t Y_{t-1} + B_t$, where $A_t = \alpha z_{t-1}^2 + (1 - \alpha)$, $B_t = \omega$
- ▣ Apply Bougerol & Picard (1992) theorem: if $\mathbb{E}[\ln A_t] < 0$ and $\mathbb{E}[\ln^+ B_t] < \infty$, the series Y_t converges in distribution

IGARCH: Strict Stationarity vs. Covariance Stationarity

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Fundamental Contrast

Property	GARCH ($\alpha + \beta < 1$)	IGARCH ($\alpha + \beta = 1$)
Strictly stationary	Yes	Yes (Nelson, 1990)
Covariance stationary	Yes	No ($\text{Var}[\varepsilon_t] = \infty$)
Unconditional variance	$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$	Does not exist (infinite)
Shock persistence	Geometric (decays)	Permanent

Practical Implication

- IGARCH is a **valid model**, despite the unit root in variance
- Conditional variance forecast: $\mathbb{E}_t[\sigma_{t+h}^2] = \sigma_t^2 + h\omega$ (linear growth)
- Does not imply process explosion: ε_t remains well-defined

Remark 1

The distinction between strict stationarity and covariance stationarity is analogous to heavy-tailed processes: a process can have a well-defined marginal distribution without having finite higher-order moments.

EWMA: Restricted GARCH for RiskMetrics

Definition 7 (Exponentially Weighted Moving Average)

The EWMA model (J.P. Morgan RiskMetrics, 1996):

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda \in (0, 1)$$

Relation to GARCH

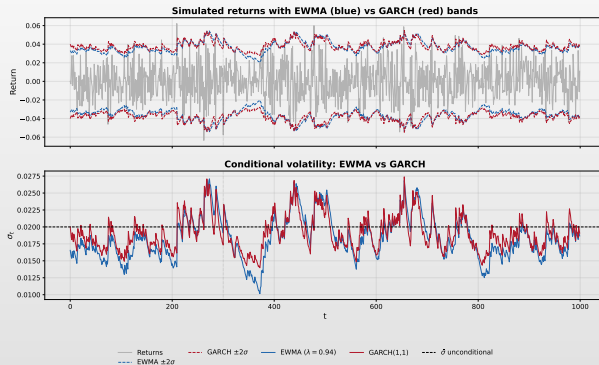
- EWMA = **IGARCH(1,1)** with $\omega = 0$: $\alpha = 1 - \lambda$, $\beta = \lambda$, $\alpha + \beta = 1$
- Industry standard: $\lambda = 0.94$ (daily data), $\lambda = 0.97$ (monthly data)
- **No estimation needed** — fixed parameter $\Rightarrow$ simple, but rigid

EWMA vs GARCH(1,1)

	EWMA	GARCH(1,1)
Estimated parameters	0 (λ fixed)	3 (ω, α, β)
Mean-reversion	No (infinite persistence)	Yes (if $\alpha + \beta < 1$)
Long-run variance	Undefined	$\omega / (1 - \alpha - \beta)$
Use case	Basel II, quick screening	Precise estimation, VaR

EWMA vs GARCH: Visual Comparison

- EWMA (blue) reacts similarly to GARCH (red), but does not revert to the unconditional level



EWMA: Interpretation and What We Observe

Reading the EWMA Series

- ▣ The EWMA volatility series closely tracks squared returns — fast reaction, no filtering
- ▣ With $\lambda = 0.94$: yesterday's weight is 6%, the rest comes from exponentially decaying history
- ▣ Visually: EWMA volatility “sticks” to spikes — rises quickly but does not revert to a fixed level

When Is EWMA Sufficient?

- ▣ **Quick screening**: no estimation required — computable immediately on any series
- ▣ **Basel II/III**: RiskMetrics EWMA remains accepted for internal VaR
- ▣ **Real-time monitoring**: simple to implement in trading systems

Limitations Visible in the Plot

- ▣ EWMA does not revert to an equilibrium level $\bar{\sigma}^2$ (has no $\omega > 0$). After crises, EWMA volatility remains elevated longer than GARCH — **overreacts on the medium term**.

CGARCH — Component GARCH (Engle & Lee, 1999)

Motivation

- GARCH(1,1) has a **single persistence component**: it cannot distinguish between short- and long-duration shocks
- CGARCH decomposes variance into a **permanent** (trend) + **transitory** component

Definition 8 (CGARCH(1,1))

$$\sigma_t^2 = q_t + \alpha(\varepsilon_{t-1}^2 - q_{t-1}) + \beta(\sigma_{t-1}^2 - q_{t-1}) \quad (\text{transitory component})$$

$$q_t = \bar{\omega} + \rho(q_{t-1} - \bar{\omega}) + \phi(\varepsilon_{t-1}^2 - \sigma_{t-1}^2) \quad (\text{permanent component})$$

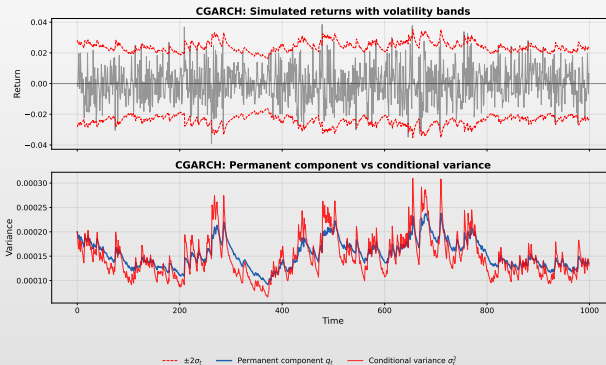
where q_t is the long-run trend, ρ controls permanent persistence ($\rho > \alpha + \beta$).

Interpretation

- q_t varies slowly (macro factors); $\sigma_t^2 - q_t$ reacts quickly to shocks
- Different half-lives: permanent $\sim$ years, transitory $\sim$ weeks. Python: `arch` with manual spec.

CGARCH Simulation: Permanent vs Transitory Component

□ q_t (blue, thick) = long-run trend; σ_t^2 (red, thin) = total volatility



CGARCH: What Do We Observe in Practice?

Interpreting the Two Components

- ▣ **Permanent component** q_t : moves slowly — macroeconomic factors (inflation, business cycle)
- ▣ **Transitory component** $\sigma_t^2 - q_t$: reacts quickly to news (earnings, Fed, geopolitics)
- ▣ Visually: q_t = smooth moving average; σ_t^2 oscillates around it

Example: S&P 500

- ▣ Permanent component rises 2005–2008, explodes 2008–2009, declines slowly afterward
- ▣ Transitory component: spikes at Flash Crash (2010), Brexit (2016), COVID (2020)
- ▣ Half-life: permanent $\approx$ 2–3 years; transitory $\approx$ 2–4 weeks

When to Use CGARCH?

- ▣ Long-horizon VaR (1 year+) — the permanent component matters for Solvency II, stress testing

FIGARCH — Fractionally Integrated GARCH

Motivation: Long Memory in Volatility

- ▣ ACF of r_t^2 decays **hyperbolically** (not exponentially) $\Rightarrow$ long memory
- ▣ GARCH: exponential decay; IGARCH: infinite persistence — neither captures long memory

Definition 9 (FIGARCH(1,d,1) — Baillie, Bollerslev & Mikkelsen (1996))

$$[1 - \beta L] \sigma_t^2 = \omega + [1 - \beta L - (1 - \phi L)(1 - L)^d] \varepsilon_t^2$$

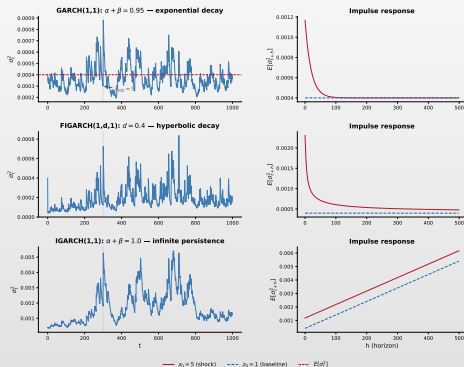
where $d \in [0, 1]$ is the fractional integration parameter: $d = 0 \Rightarrow$ GARCH, $d = 1 \Rightarrow$ IGARCH.

Key Properties

- ▣ Impulse weights $\lambda_k \sim k^{d-1}$ decay hyperbolically: old shocks matter, but less and less
- ▣ Unconditional variance: finite only if $d < 1$ (unlike IGARCH)
- ▣ Typical empirical values: $\hat{d} \approx 0.3\text{--}0.5$ for stocks (Baillie et al., 1996)

FIGARCH Simulation: GARCH vs FIGARCH vs IGARCH

- GARCH: exponential reversion; FIGARCH: hyperbolic (slow) reversion; IGARCH: infinite persistence



FIGARCH: Interpretation and Practical Applications

What Does d Mean in Practice?

- ▣ $d = 0$ (GARCH): ACF of r_t^2 decays exponentially — shocks die out quickly
- ▣ $d = 1$ (IGARCH): ACF does not decay — shocks persist infinitely
- ▣ $0 < d < 1$ (FIGARCH): ACF decays **hyperbolically** ($\sim k^{2d-1}$) — old shocks matter, but less and less

Empirical Results

- ▣ **Equities** (S&P 500, DAX): $\hat{d} \approx 0.35\text{--}0.50$ — moderate long memory
- ▣ **Forex** (EUR/USD): $\hat{d} \approx 0.40\text{--}0.60$ — higher persistence
- ▣ **Crypto** (Bitcoin): $\hat{d} \approx 0.25\text{--}0.45$ — variable over time

FIGARCH vs IGARCH: How to Choose?

- ▣ If $\hat{\alpha} + \hat{\beta} \approx 1$ in GARCH, test FIGARCH: if $\hat{d} < 1$, persistence is not infinite — just long. FIGARCH provides a more realistic picture than spurious IGARCH.

Volatility Regime Changes

Problem

GARCH assumes a single regime with constant parameters. But volatility often exhibits structural breaks (e.g., 2008 crisis, COVID-19).

Definition 10 (Markov-Switching GARCH)

Parameters (ω, α, β) switch between K regimes governed by a hidden Markov chain:

$$\sigma_t^2 = \omega_{s_t} + \alpha_{s_t} \varepsilon_{t-1}^2 + \beta_{s_t} \sigma_{t-1}^2, \quad s_t \in \{1, \dots, K\}$$

Inclan-Tiao ICSS Test

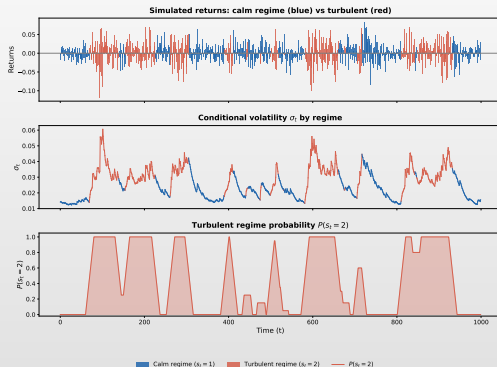
- ▣ Detects **multiple change points** in unconditional variance
- ▣ Based on cumulative sum of squared observations (CUSUM-type)
- ▣ Typical findings: S&P 500 shows breaks at 2001, 2008, 2020

Practical Implications

- ▣ If structural breaks present, GARCH persistence $(\alpha + \beta)$ is **upward biased** $\Rightarrow$ spurious IGARCH
- ▣ Solution: estimate GARCH separately per regime, or use rolling windows
- ▣ Alternative: Markov-switching GARCH with $K = 2$ (low-vol / high-vol regimes)

Markov-Switching GARCH Simulation: Volatility Regimes

- Calm regime (blue) vs turbulent regime (red) — volatility jumps between discrete levels



Regime Changes: Interpretation and Applications

What Do We See in Practice?

- ▣ Markov-switching GARCH identifies 2–3 regimes: **calm** ($\sigma \approx 0.5\text{--}1\%$) vs **turbulent** ($\sigma \approx 2\text{--}4\%$)
- ▣ The probability of the turbulent regime spikes sharply during crises (2008, 2020)

Structural Breaks Detected with ICSS on S&P 500

- ▣ **2001**: dot-com crash + 9/11; **2008**: global financial crisis; **2020**: COVID-19
- ▣ Between breaks: GARCH with different parameters is more accurate than GARCH over the entire period

Practical Implication

- ▣ If GARCH gives $\hat{\alpha} + \hat{\beta} \approx 1$, check whether it is real IGARCH or an artifact of regime changes. The ICSS test helps decide.

Outline

Motivation

Introduction to Volatility Modeling

The ARCH Model

The GARCH Model

Asymmetric GARCH Models

Model Selection and Diagnostics

Volatility Forecasting

Python Implementation

Case Study: S&P 500

Case Study: Bitcoin

AI Use Case

Summary

Quiz

From Symmetric GARCH to Asymmetric Models

What We Have Learned So Far

- ▣ GARCH(1,1) captures volatility clustering and fat tails
- ▣ IGARCH, CGARCH, FIGARCH — different types of persistence

Remark 2 (The Symmetry Limitation of Standard GARCH)

- ▣ GARCH treats shocks **symmetrically**: ε_{t-1}^2 does not distinguish the sign of ε_{t-1}
- ▣ Empirically: a -5% shock increases volatility more than $+5\%$ — **leverage effect** (Black, 1976)

Asymmetric Models Presented

- ▣ EGARCH · GJR-GARCH · TGARCH · APARCH — all distinguish the sign of the shock

Leverage Effect

Definition

Leverage effect: Negative shocks increase volatility more than positive shocks of the same magnitude.

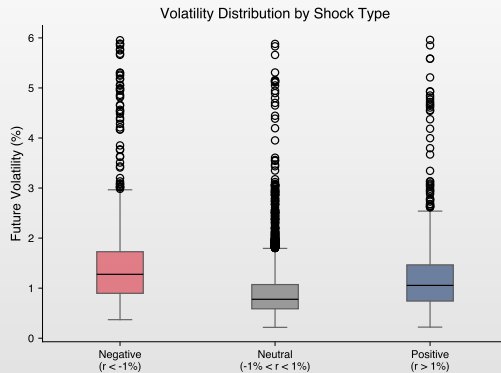
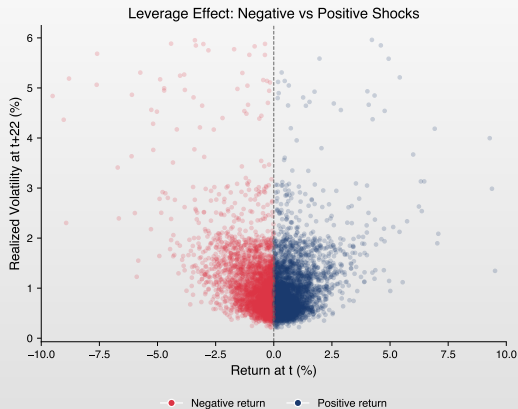
Problem with GARCH

- Standard GARCH: $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ — only ε_{t-1}^2 matters, sign is lost!
- Economic intuition: Bad news $\Rightarrow$ stock price falls $\Rightarrow$ debt/equity ratio rises $\Rightarrow$ volatility increases

Empirical Evidence

- Black (1976): first documentation of the leverage effect in equity markets
- The effect is asymmetric: 5% drops increase volatility more than 5% rises
- Solutions: EGARCH, GJR-GARCH, TGARCH — models that distinguish the sign of shocks

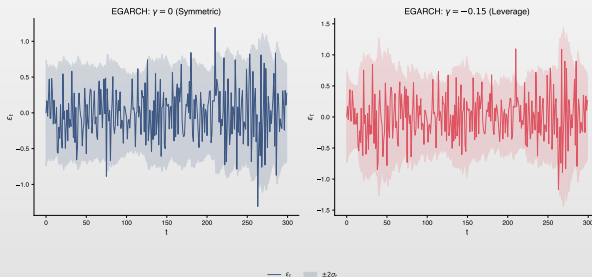
Leverage Effect



EGARCH Simulation: Symmetric vs Asymmetric

Interpretation

- When $\gamma < 0$, negative shocks increase volatility more than positive shocks



EGARCH Simulation: What Do We Observe?

Reading the Simulation Plot

- ▣ **Left panel** ($\gamma = 0$, symmetric): volatility reacts equally to positive and negative shocks — similar to standard GARCH
- ▣ **Right panel** ($\gamma < 0$, asymmetric): after **negative** shocks, volatility rises sharply and persistently; after positive shocks, the reaction is muted
- ▣ Visually: volatility spikes are **taller** and **wider** after market declines

Numerical Example: Interpreting γ

- ▣ With $\gamma = -0.15$: negative shock $z_{t-1} = -2$ contributes $0.10 \times 2 + (-0.15)(-2) = 0.50$ to $\ln(\sigma_t^2)$; positive $z_{t-1} = +2$ contributes $0.10 \times 2 + (-0.15)(+2) = -0.10$. Difference: **0.60** on log scale!

Implication for Risk Management

- ▣ EGARCH captures the fact that markets “take the elevator down and the stairs up” — volatility rises fast but falls slowly.

The EGARCH Model — Nelson (1991)

Definition 11 (EGARCH(1,1))

Exponential GARCH:

$$\ln(\sigma_t^2) = \omega + \alpha (|z_{t-1}| - \mathbb{E}[|z_{t-1}|]) + \gamma z_{t-1} + \beta \ln(\sigma_{t-1}^2)$$

where $z_t = \varepsilon_t / \sigma_t$.

EGARCH Advantages

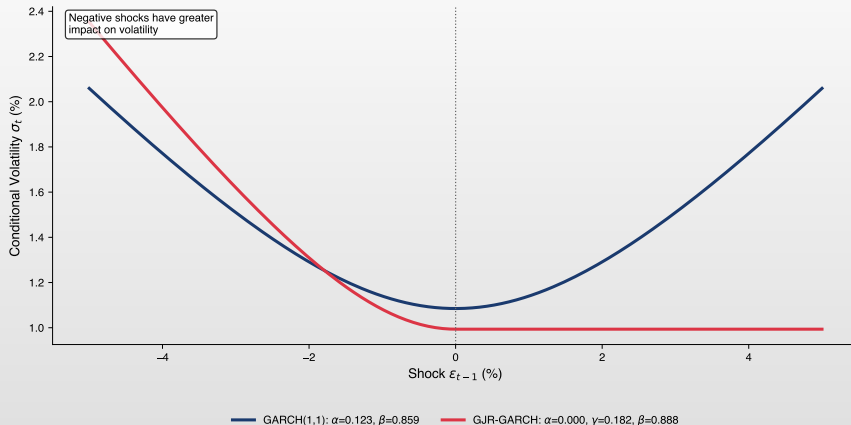
- ▣ **No non-negativity constraints required** — models $\ln(\sigma_t^2)$
- ▣ **Captures leverage effect** through parameter γ
 - ▶ $\gamma < 0$: negative shocks $\Rightarrow$ higher volatility
 - ▶ $\gamma = 0$: symmetric effect (like GARCH)
- ▣ Persistence is given by β

News Impact Curve — EGARCH

Interpretation

- ▣ **News Impact Curve:** relationship between ε_t and σ_{t+1}^2
- ▣ **GARCH:** symmetric curve (parabola)
 - ▶ Positive and negative shocks have the same impact
- ▣ **EGARCH:** asymmetric curve
 - ▶ Negative shocks have larger impact on volatility

News Impact Curve — EGARCH



EGARCH: Practical Applications

When to Use EGARCH?

- ▣ **Equity markets and indices** — pronounced leverage effect (Black, 1976)
- ▣ **Unique advantage:** no positivity constraints ($\ln \sigma_t^2$ can take any real value)
- ▣ Useful when symmetric GARCH gives $\hat{\alpha} + \hat{\beta} > 1$ (artifact of ignoring asymmetry)

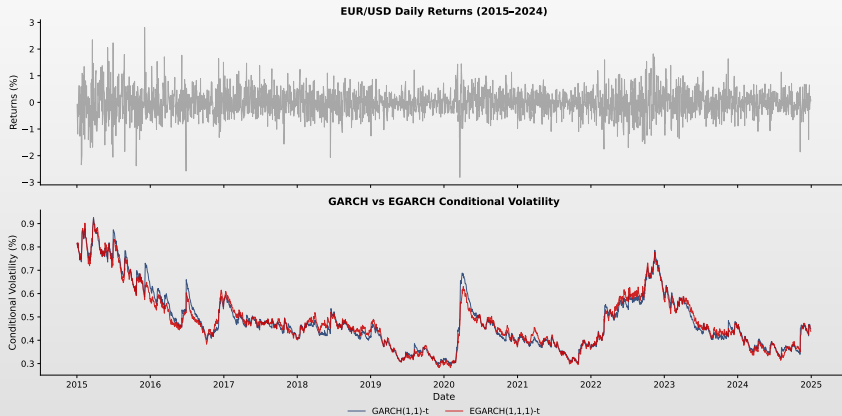
Example: S&P 500 — GARCH vs EGARCH

- ▣ GARCH(1,1): underestimates volatility after crashes (treats shocks symmetrically)
- ▣ EGARCH: volatility rises faster after drops, declines more slowly
- ▣ During COVID crisis (March 2020), EGARCH forecasts VaR 15–20% higher

Practical Rule

- ▣ Test EGARCH vs GARCH on real data: if AIC/BIC favors EGARCH, the market exhibits significant leverage.

EGARCH vs GARCH: EUR/USD — Model Comparison



The GJR-GARCH Model

Why GJR? An Alternative to EGARCH

- EGARCH models $\ln(\sigma_t^2) \Rightarrow$ hard to interpret economically. GJR-GARCH stays on the σ_t^2 scale but adds an indicator I_{t-1} that activates an extra term for negative shocks.

Definition 12 (GJR-GARCH(1,1))

Glosten, Jagannathan & Runkle (1993): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \cdot I_{t-1} + \beta \sigma_{t-1}^2$ where $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$, else 0.

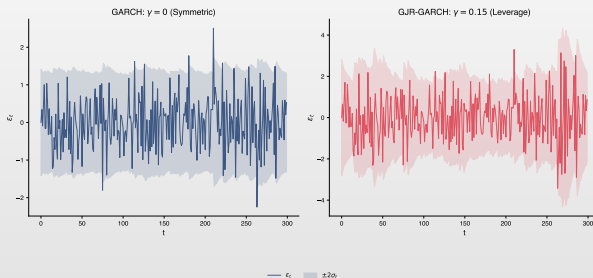
Interpretation

- Positive shocks: impact = α ; Negative shocks: impact = $\alpha + \gamma$
- Leverage effect present if $\gamma > 0$
- Stationarity: $\alpha + \gamma/2 + \beta < 1$

GJR-GARCH/TGARCH Simulation

Interpretation

- GJR-GARCH adds an indicator term to capture asymmetric response to negative shocks



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GJR-GARCH: What Do We Observe in the Simulated Series?

Reading the Simulation Plot

- ▣ The indicator $I_{t-1} = \mathbb{1}(\varepsilon_{t-1} < 0)$ “activates” only for negative shocks
- ▣ Asymmetric impact: $\varepsilon_{t-1} = -3\% \Rightarrow \text{contribution} = (\alpha + \gamma) \times 0.09 = 0.15 \times 0.09$
- ▣ Symmetric impact: $\varepsilon_{t-1} = +3\% \Rightarrow \text{contribution} = \alpha \times 0.09 = 0.05 \times 0.09$
- ▣ Visually: after down days, volatility bands widen by **3× as much**

Numerical Example: S&P 500

With $\alpha = 0.05$, $\gamma = 0.10$, $\beta = 0.88$, $\omega = 0.00001$:

- ▣ Good day (+2%): $\Delta\sigma_t^2 \approx 0.05 \times 0.04\% = 0.002\%$
- ▣ Bad day (−2%): $\Delta\sigma_t^2 \approx (0.05 + 0.10) \times 0.04\% = 0.006\%$ — **triple!**

GJR vs EGARCH: When to Choose GJR?

- ▣ GJR operates on the σ_t^2 scale $\Rightarrow$ coefficients are directly interpretable as marginal impacts.
EGARCH operates on $\ln(\sigma_t^2)$ $\Rightarrow$ harder to read economically, but no positivity constraints.

The APARCH Model — Ding, Granger & Engle (1993)

Definition 13 (APARCH(1,1))

Asymmetric Power ARCH — the unifying model of the GARCH family:

$$\sigma_t^\delta = \omega + \alpha_1(|\varepsilon_{t-1}| - \gamma_1\varepsilon_{t-1})^\delta + \beta_1\sigma_{t-1}^\delta$$

where $\delta > 0$ is the power parameter and $|\gamma_1| < 1$ captures asymmetry.

Special Cases

- ▣ $\delta = 2, \gamma_1 = 0$: **GARCH(1,1)** standard
- ▣ $\delta = 2, \gamma_1 \neq 0$: **GJR-GARCH**
- ▣ $\delta = 1, \gamma_1 \neq 0$: **TGARCH** (Zakoian, 1994)
- ▣ $\delta = 1, \gamma_1 = 0$: **AVGARCH** (Taylor, 1986)

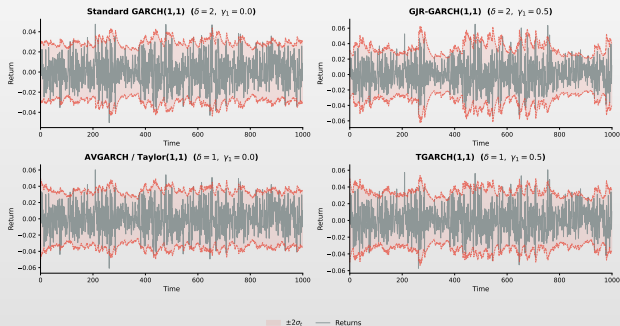
Why APARCH Matters

- ▣ Estimating δ from data (not fixed at 1 or 2) can **improve forecasts**
- ▣ Empirical studies: $\hat{\delta} \approx 1.0\text{--}1.5$ for equities (Ding et al., 1993)
- ▣ Provides a **unifying framework** for comparing asymmetric models

APARCH Simulation: Special Cases

- APARCH unifies the entire GARCH family: δ and γ control power and asymmetry

APARCH Model: Effect of δ and γ_1 Parameters



APARCH: Interpretation and Practical Applications

What Does the Estimate of δ Tell Us?

- ▣ $\hat{\delta} \approx 2$: the market behaves like standard GARCH (variance is the natural measure)
- ▣ $\hat{\delta} \approx 1$: modeling standard deviation (σ_t) is more appropriate — like TGARCH
- ▣ $\hat{\delta} < 1$: absolute returns have stronger structure than squares

Typical Empirical Results

- ▣ **Equities**: $\hat{\delta} \approx 1.0\text{--}1.5$, $\hat{\gamma} \approx 0.3\text{--}0.8$ (strong leverage)
- ▣ **Forex**: $\hat{\delta} \approx 1.5\text{--}2.0$, $\hat{\gamma} \approx 0$ (symmetric, no leverage)
- ▣ **Crypto**: $\hat{\delta} \approx 1.0\text{--}1.3$, $\hat{\gamma}$ variable (depends on the period)

When to Use APARCH?

- ▣ When we do not know a priori what power (δ) or what level of asymmetry (γ) is optimal — APARCH lets the data decide. Useful as a reference model for testing restrictions (LR test vs GARCH, GJR, etc.).

Stationarity of Asymmetric Models

Stationarity Conditions — Asymmetric Models

Model	Condition	Notes
GARCH(1,1)	$\alpha + \beta < 1$	Covariance-stationarity
GJR-GARCH	$\alpha + \gamma/2 + \beta < 1$	$\gamma/2$ because $P(\varepsilon < 0) = 1/2$
EGARCH(1,1)	$ \beta < 1$	$\ln \sigma_t^2$ is AR(1)-like
TGARCH(1,1)	$(\alpha^+ + \alpha^-)/2 + \beta < 1$	Mean of positive/negative impact
APARCH(1,1)	$\alpha_1 \mathbb{E}[(z - \gamma z)^\delta] + \beta_1 < 1$	Depends on distribution of z_t

Proposition 1 (EGARCH Stationarity (Nelson, 1991))

- EGARCH models $\ln \sigma_t^2$ as an AR(1) process with innovations $g(z_{t-1})$
- $|\beta| < 1$: **necessary and sufficient** for strict stationarity
- **Note:** do not confuse with $\alpha + \beta < 1$ from GARCH — α in EGARCH has a different role

TGARCH — Threshold GARCH

Why TGARCH? Direct Modeling of Standard Deviation

- GARCH/EGARCH/GJR model σ_t^2 . TGARCH models σ_t directly, allowing separate coefficients α^+ , α^- for positive and negative shocks — more intuitive interpretation.

Definition 14 (TGARCH(1,1))

Zakoian (1994) models standard deviation: $\sigma_t = \omega + \alpha^+ \varepsilon_{t-1}^+ + \alpha^- \varepsilon_{t-1}^- + \beta \sigma_{t-1}$

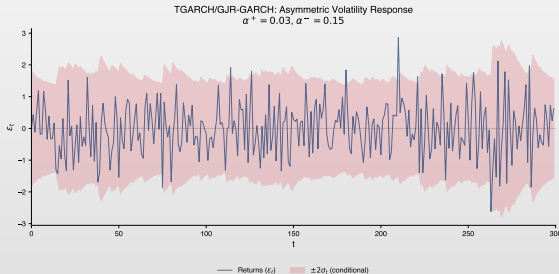
Comparison of Asymmetric Models

Model	Specification	Leverage
GARCH	σ_t^2	No
EGARCH	$\ln(\sigma_t^2)$	Yes ($\gamma < 0$)
GJR-GARCH	σ_t^2 with indicator	Yes ($\gamma > 0$)
TGARCH	σ_t	Yes ($\alpha^- > \alpha^+$)
APARCH	σ_t^δ	Yes ($\gamma_i \neq 0$)

TGARCH Simulation: Asymmetric Volatility Response

Interpretation

- TGARCH with $\alpha^+ = 0.03$ and $\alpha^- = 0.15 \Rightarrow$ negative shocks amplify volatility by $5\times$
- Volatility bands $\pm 2\sigma$ widen asymmetrically during crisis periods



TGARCH: What Do We Observe in the Simulated Series?

Reading the Simulation Plot

- ▣ With $\alpha^- = 0.15$ and $\alpha^+ = 0.03$, a negative shock of -3% increases σ_t by $0.15 \times 3 = 0.45$
- ▣ The same positive shock of $+3\%$ increases σ_t by only $0.03 \times 3 = 0.09$ — **5× less**
- ▣ Visually: the $\pm 2\sigma$ bands widen abruptly and **asymmetrically** after market drops

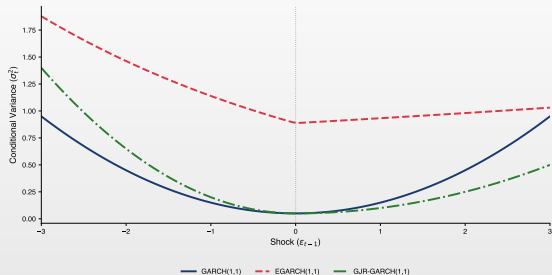
When to Use TGARCH?

- ▣ When we want to model σ_t directly (not σ_t^2 like GJR, nor $\ln \sigma_t^2$ like EGARCH)
- ▣ Markets with strong leverage: individual stocks, equity indices
- ▣ **Advantage:** natural interpretation — coefficients act on the standard deviation scale

TGARCH vs GJR-GARCH

- ▣ Both capture asymmetry, but on different scales (σ_t vs σ_t^2). Empirically, differences are small; TGARCH is preferred in risk management (VaR uses σ_t directly).

News Impact Curves Comparison



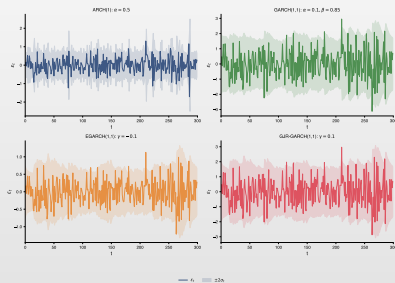
Interpretation

- **Standard GARCH:** symmetric
 - ▶ Treats positive and negative shocks identically
- **EGARCH and GJR-GARCH:** capture asymmetry
 - ▶ Leverage effect: negative shocks $\Rightarrow$ larger impact

GARCH Family Comparison

Interpretation

- All models capture volatility clustering, but differ in how they model asymmetry



When to Choose ARCH, GARCH, EGARCH or GJR-GARCH?

Selection Guide

- ▣ **ARCH(q)**: conceptually simple, but requires many lags $\Rightarrow$ **rarely used in practice**
- ▣ **GARCH(1,1)**: **the standard benchmark** — captures persistence with only 3 parameters; sufficient in $\sim 90\%$ of cases
- ▣ **EGARCH**: models $\ln(\sigma_t^2) \Rightarrow$ **no positivity constraints**; captures leverage through γ ; **preferred for equities**
- ▣ **GJR-GARCH**: indicator variable $I(\varepsilon_{t-1} < 0) \Rightarrow$ **direct interpretation of asymmetric impact**; equivalent to EGARCH

Practical Selection Rules

- ▣ **Step 1**: Start with GARCH(1,1) — sufficient for most series
- ▣ **Step 2**: Check for leverage effect (sign bias test, news impact curve)
- ▣ **Step 3**: If leverage significant $\Rightarrow$ EGARCH or GJR-GARCH
- ▣ **Step 4**: Compare with AIC/BIC; choose Student-t distribution

Models are not a checklist — they are a set of data-driven choices.

Asset Class Selection Guide

Which model for which market?

Asset Class	Recommended Model	Why?
Indices (S&P 500, DAX)	EGARCH-t / GJR-t	Strong leverage
Individual stocks	GJR-GARCH-t	Leverage + interpretability
Forex (EUR/USD)	GARCH(1,1)-t / FIGARCH	High persistence, no leverage
Crypto (Bitcoin)	EGARCH-t / GJR-t	Leverage + very fat tails
Commodities (oil, gold)	GARCH(1,1)-t	Moderate, no clear asymmetry
Interest rates	IGARCH / CGARCH	Extreme persistence, regimes

Practical Selection Rules

- ▣ **Step 1:** GARCH(1,1)-t → **Step 2:** sign bias test → **Step 3:** if leverage ⇒ EGARCH/GJR
- ▣ **Step 4:** if $\hat{\alpha} + \hat{\beta} \approx 1$ ⇒ FIGARCH or Markov-switching
- ▣ **Step 5:** compare AIC/BIC + Diebold-Mariano out-of-sample

GARCH-in-Mean (GARCH-M) — Engle, Lilien & Robins (1987)

Definition 15 (GARCH-M)

- ▣ **Model:** Volatility enters directly into the mean equation:

$$r_t = \mu + \delta \cdot g(\sigma_t^2) + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t$$
$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- ▣ **Function** g : can be σ_t^2 , σ_t , or $\ln(\sigma_t^2)$

Economic Interpretation

- ▣ $\delta > 0$: **risk premium** $\Rightarrow$ higher returns when volatility is high
- ▣ Formalizes the risk-return relationship (CAPM, Merton ICAPM); test $H_0 : \delta = 0$

Typical Example: Equities

- ▣ $r_t = 0.02 + \underbrace{0.15}_{\delta} \cdot \sigma_t + \varepsilon_t \quad \Rightarrow \text{At } \sigma_t = 2\%: \mathbb{E}[r_t] = 0.023 \text{ (0.3\% premium)}$

GARCH-M: Alternative Specifications

Common Specifications

Risk premium can enter in different forms: (1) $r_t = \mu + \lambda\sigma_t + \varepsilon_t$; (2) $r_t = \mu + \lambda\sigma_t^2 + \varepsilon_t$; (3) $r_t = \mu + \lambda \ln(\sigma_t^2) + \varepsilon_t$

Typical Results for Equity Markets

- ▣ Estimated λ often positive but small (0.01–0.10)
- ▣ Significance varies across markets and periods
- ▣ Variance specification yields larger λ estimates

Remark 3

GARCH-M is used in asset pricing, portfolio optimization, and CAPM testing.

Outline

Motivation

Introduction to Volatility Modeling

The ARCH Model

The GARCH Model

Asymmetric GARCH Models

Model Selection and Diagnostics

Volatility Forecasting

Python Implementation

Case Study: S&P 500

Case Study: Bitcoin

AI Use Case

Summary

Quiz

Complete Workflow: From Data to Validated VaR

Practical 8-Step Algorithm

1. **Estimate the mean equation:** ARMA or $r_t = \mu + \varepsilon_t$ (check ACF of returns)
2. **Test for ARCH effects:** ARCH-LM test + Ljung-Box on $\hat{\varepsilon}_t^2$
3. **Estimate the GARCH family:** GARCH(1,1), EGARCH(1,1), GJR-GARCH(1,1)
4. **Compare innovation distributions:** Normal vs Student-t vs GED
5. **Check standardized residuals:** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ i.i.d. (Ljung-Box, ARCH-LM residual, Q-Q plot)
6. **Compare models with AIC/BIC:** lowest AIC/BIC = best trade-off
7. **Forecast volatility:** $\hat{\sigma}_{T+h}^2$ with rolling window or recursive
8. **Validate VaR:** backtesting with Kupiec (coverage) and Christoffersen (independence)

Remember

- This is **exactly** the workflow you will follow in the S&P 500 and Bitcoin case studies
- **Even though returns are hard to predict, their volatility is often predictable**

Order Selection

Information Criteria

- ▣ **AIC** = $-2\ell + 2k$
- ▣ **BIC** = $-2\ell + k \ln(T)$
- ▣ **HQIC** = $-2\ell + 2k \ln(\ln(T))$

where ℓ = maximized log-likelihood, k = number of parameters.

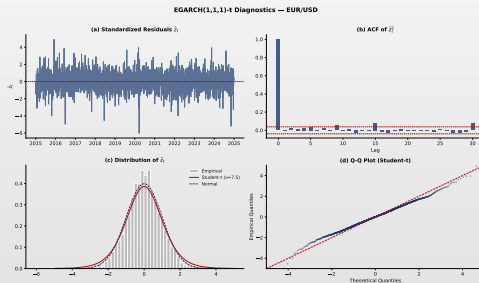
Practical Recommendations

- ▣ GARCH(1,1) is sufficient in **90% of cases**
- ▣ Check if asymmetric model significantly improves fit
- ▣ Choose innovation distribution that minimizes AIC/BIC

Diagnostic Example — EUR/USD EGARCH(1,1,1)-t

Verification

- Standardized residuals should be i.i.d. with no residual ARCH effects



GARCH Model Diagnostics

Standardized Residuals

$$\hat{z}_t = \frac{\hat{\varepsilon}_t}{\hat{\sigma}_t}$$

If the model is correctly specified, $\hat{z}_t$ should be i.i.d.(0,1).

Diagnostic Checks

1. **Ljung-Box on $\hat{z}_t$** : check absence of autocorrelation in mean
2. **Ljung-Box on $\hat{z}_t^2$** : check absence of residual ARCH effects
3. **ARCH-LM test on $\hat{z}_t$** : confirm absence of heteroskedasticity
4. **Histogram + QQ-plot**: verify assumed distribution

Sign-Bias Test — Engle & Ng (1993)

Motivation

- ▣ Tests whether the GARCH model **correctly captures asymmetry** in volatility
- ▣ If the test rejects, the symmetric model is inadequate $\Rightarrow$ need EGARCH or GJR

Definition 16 (Sign-Bias Regressions)

Let $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ and $S_t^- = 1(\hat{\varepsilon}_t < 0)$. Estimate:

$$\hat{z}_t^2 = c_0 + c_1 S_{t-1}^- + c_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + c_3 (1 - S_{t-1}^-) \hat{\varepsilon}_{t-1} + u_t$$

- ▣ **Sign-bias:** $H_0 : c_1 = 0$; **Neg. size-bias:** $H_0 : c_2 = 0$; **Pos. size-bias:** $H_0 : c_3 = 0$
- ▣ **Joint test:** $H_0 : c_1 = c_2 = c_3 = 0$, statistic $TR^2 \sim \chi^2(3)$

Practical Interpretation

- ▣ Rejection $\Rightarrow$ model does not capture leverage $\Rightarrow$ switch to EGARCH/GJR
- ▣ Acceptance on asymmetric model $\Rightarrow$ asymmetry well specified

Nyblom Stability Test (1989)

Motivation

- ▣ Tests whether GARCH parameters are **constant over the entire period**
- ▣ Parameter instability distorts persistence: $\hat{\alpha} + \hat{\beta}$ artificially inflated

Definition 17 (Nyblom Test)

H_0 : $\theta = (\omega, \alpha, \beta, \dots)$ constant; H_1 : θ_t follows a random walk.

$$L_c = \frac{1}{T^2} \sum_{t=1}^T S_t' \hat{V}^{-1} S_t, \quad S_t = \sum_{s=1}^t \frac{\partial \ell_s}{\partial \theta} \Big|_{\hat{\theta}}$$

S_t = partial sum of scores; $\hat{V}$ = estimated information matrix.

Practical Usage

- ▣ Python: `result.diagnostic_summary()` (arch package)
- ▣ Rejection $\Rightarrow$ rolling window or regime-switching GARCH

Estimated Information Matrix

What is the information matrix?

- ▣ **Score:** $s_t(\theta) = \nabla_{\theta} \ell_t$ — derivative of log-likelihood w.r.t. parameters
- ▣ **Fisher information:** $\mathcal{I}(\theta) = \mathbb{E}[s_t s_t']$ — measures how much data “inform” about parameters

Estimating the Information Matrix

- ▣ **OPG** (Outer Product of Gradients): $\hat{V} = \frac{1}{T} \sum_{t=1}^T s_t s_t'$
- ▣ **Hessian:** $\hat{V} = -\frac{1}{T} \sum_{t=1}^T \nabla_{\theta}^2 \ell_t$
- ▣ **Sandwich** (robust, Bollerslev-Wooldridge): $\hat{V} = H^{-1} G H^{-1}$, H = Hessian, G = OPG

Why does it matter?

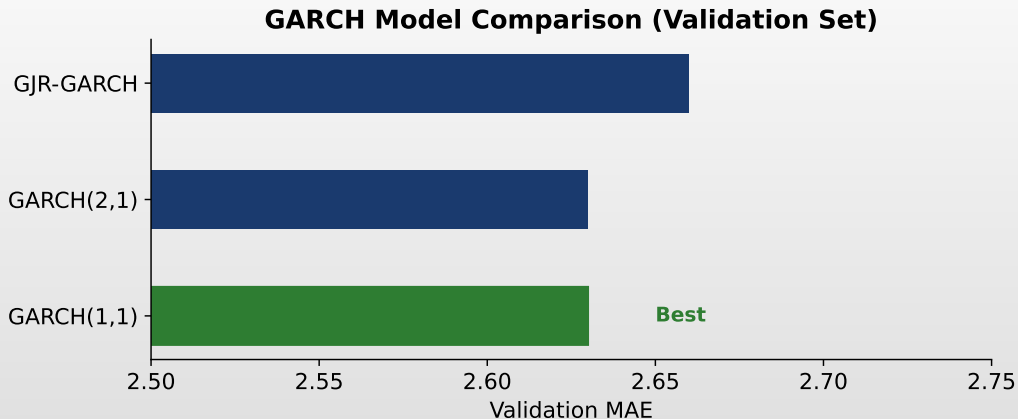
Standard errors: $se(\hat{\theta}_j) = \sqrt{[\hat{V}^{-1}]_{jj}/T}$. The sandwich version is robust to distributional misspecification.

GARCH Model Comparison: Validation

Interpretation

- ▣ GARCH(1,1) achieves the lowest MAE on the validation set
 - ▶ More parsimonious and stable than higher-order models
- ▣ GARCH(2,1) and GJR-GARCH: similar performance, but more parameters
- ▣ **Conclusion:** simplicity wins $\Rightarrow$ GARCH(1,1) is hard to beat

GARCH Model Comparison: Validation



Diebold-Mariano Test

Motivation

Given two competing volatility models, which forecasts better?

Definition 18 (Diebold-Mariano (1995))

Define the loss differential: $d_t = L(e_{1t}) - L(e_{2t})$ where $e_{it} = \sigma_t^2 - \hat{\sigma}_{it}^2$ and $L(\cdot)$ is a loss function (e.g., $L(e) = e^2$).

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}} \xrightarrow{d} N(0, 1) \quad \text{under } H_0 : \mathbb{E}[d_t] = 0$$

Practical Notes

- H_0 : equal predictive accuracy; rejection $\Rightarrow$ one model is significantly better
- Common loss functions: MSE (e^2), MAE ($|e|$), QLIKE ($\log \hat{\sigma}^2 + \sigma^2/\hat{\sigma}^2$)
- QLIKE is robust to noisy volatility proxies (Patton, 2011)
- Use HAC standard errors (Newey-West) if d_t is autocorrelated

Example

EGARCH-t vs GARCH-N on S&P 500: DM test with QLIKE loss typically rejects H_0 in favor of EGARCH-t.

Loss Functions for Volatility Forecast Evaluation

Why does the loss function matter?

- True volatility σ_t^2 is **unobservable** — we use proxies (r_t^2 , RV_t)
- The loss function must be **consistent**: the ranking does not depend on the proxy (Patton, 2011)

Definition 19 (Standard Loss Functions)

$$L_{\text{MSE}} = (\sigma_t^2 - \hat{\sigma}_t^2)^2 \quad (\text{sensitive to extremes})$$

$$L_{\text{QLIKE}} = \ln \hat{\sigma}_t^2 + \sigma_t^2 / \hat{\sigma}_t^2 \quad (\text{robust, consistent — recommended})$$

$$L_{\text{MAE}} = |\sigma_t - \hat{\sigma}_t| \quad (\text{not consistent for } \sigma^2)$$

Recommendation

- **QLIKE**: robust to noisy proxy; available in DM and MCS

Model Confidence Set — Hansen, Lunde & Nason (2011)

Motivation: from pairwise to multiple comparison

- ▣ Diebold-Mariano compares **2 models**; what about 5, 10, or 20 candidates?
- ▣ MCS: identifies the **set of statistically equivalent models** (“the best”)

Definition 20 (Model Confidence Set (MCS))

Starting from $\mathcal{M}_0 = \{M_1, \dots, M_m\}$ (all candidate models):

1. Test H_0 : all models in the set have equal predictive accuracy
2. If rejected: eliminate the weakest model ($t_{\max}$ or T_R)
3. Repeat until H_0 is no longer rejected $\Rightarrow \hat{\mathcal{M}}_{1-\alpha}^*$

Advantage over Diebold-Mariano

- ▣ Controls **multiple testing error** (familywise error rate)
- ▣ Each model receives an **MCS p-value**; available: `arch.bootstrap.MCS`

Mincer-Zarnowitz Regression

Forecast Efficiency Test

Regress realized volatility on forecasted volatility:

$$\sigma_{t,\text{realized}}^2 = a + b \hat{\sigma}_t^2 + u_t$$

Definition 21 (Hypotheses)

$H_0 : a = 0, b = 1$ (unbiased, efficient forecast)

Test via F-test or Wald test. Rejection $\Rightarrow$ forecast is biased or inefficient.

Interpretation

- ▣ R^2 measures fraction of variance explained by the forecast
- ▣ $b > 1$: forecasts underestimate volatility; $b < 1$: overestimate
- ▣ $a > 0$: systematic upward bias in realized volatility

Caveat

Realized variance (r_t^2) is a noisy proxy for $\sigma_t^2 \Rightarrow R^2$ typically low (< 0.10). Prefer RV_t when available.

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Summary

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From Modeling to Forecasting and Risk Management

What we have built so far

- ▣ Estimated GARCH models (symmetric and asymmetric)
- ▣ Validated with diagnostic tests (Ljung-Box, ARCH-LM, Sign-bias, Nyblom)
- ▣ Selected the best model (AIC/BIC, Diebold-Mariano, MCS)

Next step: what is the model for?

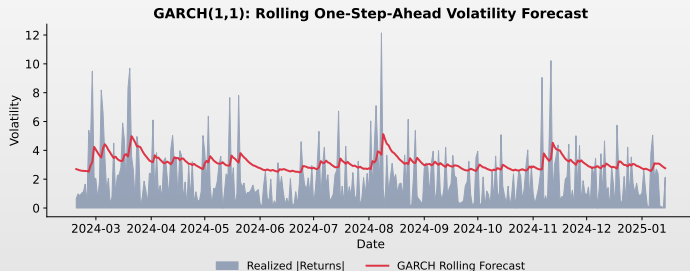
- ▣ **Volatility forecasting:** $\hat{\sigma}_{T+h}^2$ — convergence to $\bar{\sigma}^2$
- ▣ **Value at Risk (VaR):** maximum loss at a given probability
- ▣ **Expected Shortfall (ES):** average loss beyond VaR (Basel III/IV)
- ▣ **Backtesting:** validating the VaR model on historical data

Section Structure

Forecast → VaR/ES → Backtesting (Kupiec, Christoffersen) → Applied case studies

Volatility Forecast — Visualization

- Forecast converges exponentially to $\bar{\sigma}^2$; speed depends on $\alpha + \beta$
- The closer $\alpha + \beta$ is to 1, the slower the convergence



Forecasting with GARCH(1,1)

One-Step-Ahead Forecast

$$\hat{\sigma}_{T+1}^2 = \omega + \alpha \varepsilon_T^2 + \beta \sigma_T^2$$

Multi-Step Forecast

For $h > 1$: $\mathbb{E}_T[\sigma_{T+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{T+1}^2 - \bar{\sigma}^2)$ where $\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$ = unconditional variance.

Convergence

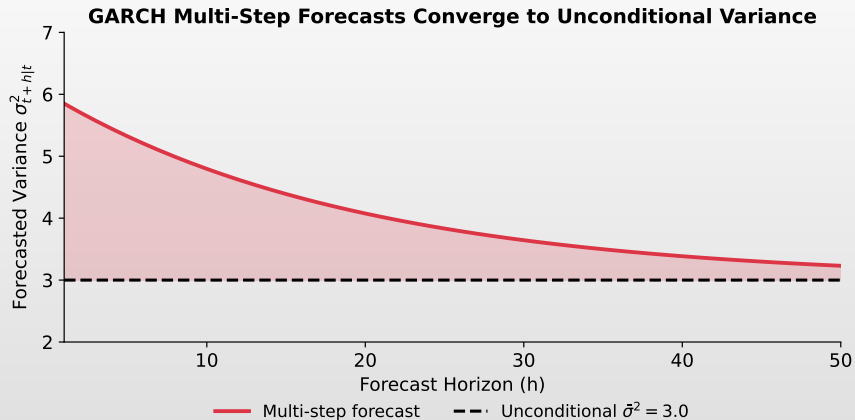
$\lim_{h \rightarrow \infty} \mathbb{E}_T[\sigma_{T+h}^2] = \bar{\sigma}^2$ — forecast converges to unconditional variance!

GARCH Forecast Convergence to Unconditional Variance

Interpretation

- ▣ Multi-step forecast converges exponentially to $\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$
- ▣ The closer $\alpha + \beta$ is to 1, the slower the convergence
 - ▶ S&P 500: $\alpha + \beta \approx 0.99 \Rightarrow$ convergence in ~ 50 days
 - ▶ Bitcoin: $\alpha + \beta \approx 0.95 \Rightarrow$ faster convergence

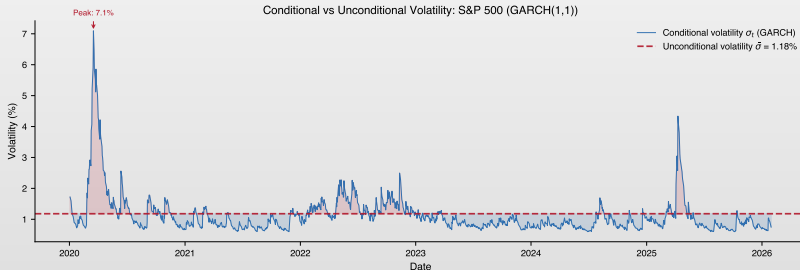
GARCH Forecast Convergence to Unconditional Variance



Conditional vs Unconditional Volatility

Interpretation

- Conditional volatility σ_t (GARCH): time-varying, reacts to news
- Unconditional volatility $\bar{\sigma} = \sqrt{\omega/(1 - \alpha - \beta)}$: long-run constant level
- Red zones: $\sigma_t > \bar{\sigma}$ (high-stress periods); blue zones: $\sigma_t < \bar{\sigma}$ (calm periods)



Realized Volatility and HAR-RV

Realized Volatility (RV)

With intraday returns $r_{t,i}$ at frequency Δ (M observations per day):

$$RV_t = \sum_{i=1}^M r_{t,i}^2$$

RV_t is a **model-free, observable** measure of daily variance (vs. GARCH: latent).

Definition 22 (HAR-RV Model — Corsi (2009))

$$RV_t = \beta_0 + \beta_D RV_{t-1} + \beta_W RV_{t-1}^{(w)} + \beta_M RV_{t-1}^{(m)} + u_t$$

where $RV^{(w)} = \frac{1}{5} \sum_{j=1}^5 RV_{t-j}$ (weekly) and $RV^{(m)} = \frac{1}{22} \sum_{j=1}^{22} RV_{t-j}$ (monthly).

Key Advantages

- Simple OLS regression that captures **long memory** in volatility
- Heterogeneous traders (daily/weekly/monthly) explain multi-scale dynamics
- Often outperforms GARCH when intraday data available (5-min returns, $M \approx 78/\text{day}$)
- Caveat: microstructure noise at very high frequency $\Rightarrow$ use kernel estimators

Realized GARCH

Motivation

Standard GARCH treats volatility as latent. With high-frequency data, we can observe it via RV_t .

Definition 23 (Realized GARCH — Hansen, Huang & Shek (2012))

Return equation: $r_t = \mu + \sigma_t z_t$, $z_t \sim D(0, 1)$

GARCH equation: $\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \gamma \log RV_{t-1}$

Measurement equation: $\log RV_t = \xi + \varphi \log \sigma_t^2 + \tau(z_t) + u_t$

Benefits

- ▣ **Joint model:** links returns, latent volatility, and realized measures
- ▣ Leverage function $\tau(z_t)$ captures asymmetry directly
- ▣ Typically **more accurate** forecasts than standard GARCH (uses richer information set)

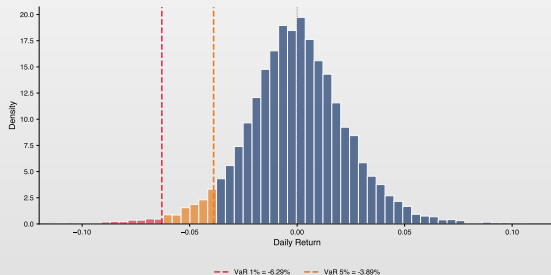
Comparison: GARCH vs HAR-RV vs Realized GARCH

- ▣ **GARCH:** no intraday data needed; most widely used
- ▣ **HAR-RV:** simple, model-free; requires 5-min data
- ▣ **Realized GARCH:** best of both worlds; requires 5-min data + parametric structure

VaR and ES: Graphical Illustration

Interpretation

- ▣ VaR 1% = loss exceeded only in 1% of cases
- ▣ Red area = extreme losses (beyond VaR)



Applications of Volatility Forecasting

Value at Risk (VaR)

$$\text{VaR}_\alpha = z_\alpha \cdot \sigma_{T+1}$$

Maximum loss with probability $1 - \alpha$.

Expected Shortfall (ES)

$$\text{ES}_\alpha = \mathbb{E}[-r | r < -\text{VaR}_\alpha]$$

Average loss when VaR is exceeded.

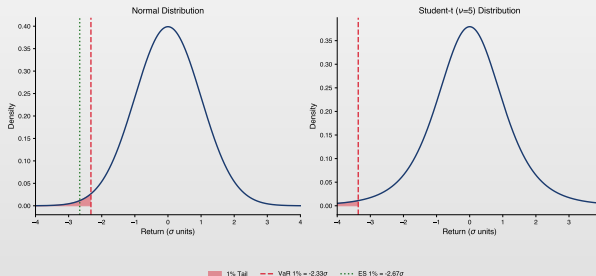
Other Applications

- Option pricing
- Dynamic hedging
- Portfolio allocation
- Stress testing

VaR vs Expected Shortfall: Normal vs Student-t

Interpretation

- ES measures average loss when VaR is exceeded
- Student-t: VaR and ES are larger than under normal distribution



Value at Risk — Numerical Example

VaR Calculation

Portfolio: **1,000,000 EUR**, forecasted volatility $\hat{\sigma}_{T+1} = 1.5\%$

VaR with Normal Distribution

Level	z_{α}	VaR (%)	VaR (EUR)
5% (1 day)	1.645	2.47%	24,675
1% (1 day)	2.326	3.49%	34,890

Scaling for Longer Periods

- ▣ $\text{VaR}_{h \text{ days}} = \text{VaR}_{1 \text{ day}} \cdot \sqrt{h}$ — assumes i.i.d. returns
- ▣ **Warning:** Under GARCH models, returns are **not** i.i.d. (they are conditionally heteroskedastic) $\Rightarrow$ the $\sqrt{h}$ rule is only an **approximation**; for multi-period VaR, use iterated GARCH forecasts

Value at Risk — Student-t Distribution

Why Student-t?

Normal distribution **underestimates** tail risk. Student-t with ν degrees of freedom better captures fat tails (kurtosis > 3).

VaR 1% (1 day) Comparison: $\sigma = 1.5\%$, Portfolio = 1M EUR

Distribution	Quantile	VaR (EUR)
Normal	2.326	34,890
Student-t ($\nu = 6$)	3.143	47,145
Student-t ($\nu = 4$)	3.747	56,205

Observation

With $\nu = 6$ (typical for stocks), VaR is **35% higher** than normal!

VaR — Complete Example with GARCH

VaR Calculation Procedure

1. Estimate GARCH(1,1) model with Student-t distribution
2. Obtain volatility forecast: $\hat{\sigma}_{T+1}$
3. Calculate VaR: $\text{VaR}_\alpha = t_\alpha(\nu) \cdot \hat{\sigma}_{T+1} \cdot \sqrt{\frac{\nu-2}{\nu}}$

Example: S&P 500

- ▣ Estimated parameters: $\alpha = 0.088$, $\beta = 0.900$, $\nu = 6.4$
- ▣ Forecasted volatility: $\hat{\sigma}_{T+1} = 1.2\%$
- ▣ Portfolio: 10,000,000 EUR

VaR 1% (1 day): $\text{VaR} = 3.05 \times 0.012 \times 10,000,000 = 366,000 \text{ EUR}$

What is VaR Backtesting?

Definition

- **Backtesting** = ex-post verification of VaR model quality
- Compares realized losses with the forecasted VaR threshold
 - ▶ A **violation** occurs when $r_t < -\text{VaR}_t$

Backtesting Principle

- Violation indicator: $I_t = 1(r_t < -\text{VaR}_{\alpha,t})$
- For a correctly specified model at level α :
 - ▶ Frequency: $\hat{p} = \frac{1}{T} \sum I_t \approx \alpha$; violations **independent**
- VaR 1% over 250 days $\Rightarrow$ expect ~ 2.5 violations/year

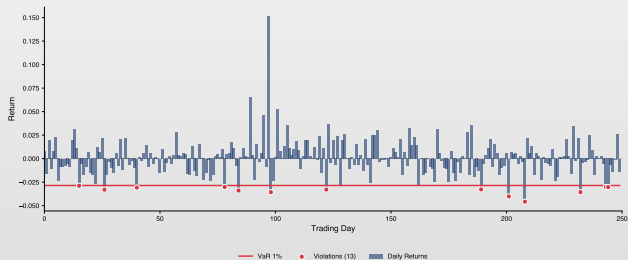
Importance

- Regulatory requirement under **Basel III/IV** for banks: backtesting is mandatory

VaR Backtesting: Visual Overview

Interpretation

- Red line: VaR 1% threshold estimated with GARCH(1,1)
- Red dots: 13 violations out of 250 days ($\hat{p} = 5.2\%$)
 - ▶ **Basel red zone** $\Rightarrow$ model significantly underestimates risk
 - ▶ Solutions: Student-t distribution, EGARCH model, or more conservative VaR level



VaR Backtesting: Basel Traffic Light

Basel III/IV Traffic Light Zones

Zone	Violations/250 days	Interpretation	Penalty
Green	0–4	Model acceptable	No penalty
Yellow	5–9	Needs investigation	Factor k increases
Red	≥ 10	Model inadequate	Maximum penalty

Practical Example

- Portfolio with VaR 1%: 250 days of backtesting
- 3 violations $\Rightarrow$ **Green zone** $\Rightarrow$ model acceptable
- 7 violations $\Rightarrow$ **Yellow zone** $\Rightarrow$ revision needed
- 13 violations $\Rightarrow$ **Red zone** $\Rightarrow$ model rejected

Rolling Window VaR Methodology

Rolling Window Concept

- ▣ A **rolling window** of fixed size W (e.g., 500 days) moves day by day
- ▣ At each step t : re-estimate GARCH on $[t - W, t - 1]$, forecast $\hat{\sigma}_{t|t-1}$, compute VaR_t

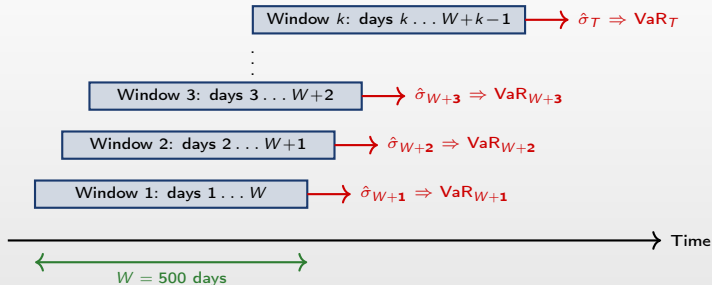
Step-by-Step Procedure (for each day $t = W + 1, \dots, T$)

1. Estimate GARCH on $\{r_{t-W}, \dots, r_{t-1}\} \Rightarrow$ parameters $\hat{\omega}, \hat{\alpha}, \hat{\beta}, \hat{\nu}$
2. Forecast: $\hat{\sigma}_{t|t-1}^2 = \hat{\omega} + \hat{\alpha}r_{t-1}^2 + \hat{\beta}\hat{\sigma}_{t-1}^2$
3. Compute: $\text{VaR}_{\alpha,t} = -t_{\alpha}(\hat{\nu}) \cdot \sqrt{\frac{\hat{\nu}-2}{\hat{\nu}}} \cdot \hat{\sigma}_{t|t-1}$
4. Check violation: $I_t = 1(r_t < -\text{VaR}_{\alpha,t})$

Why Rolling and Not Expanding?

- ▣ Fixed window: parameters reflect the **current regime** of volatility
- ▣ Old data ($> W$ days) may be irrelevant (structural changes, crises)

Rolling Window VaR: Procedure Diagram



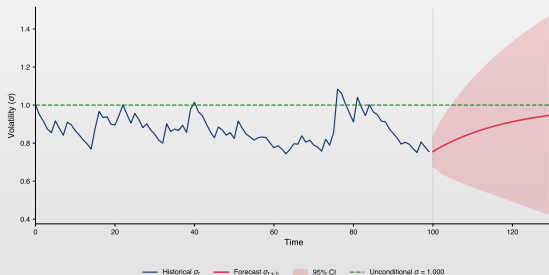
Result

- We obtain the series $\{\text{VaR}_{\alpha,t}\}_{t=W+1}^T \Rightarrow$ a **different** threshold every day
- VaR adapts to the current regime: increases during volatile periods, decreases during calm ones
- Compare r_t with $-\text{VaR}_{\alpha,t}$ to identify violations

Volatility Forecast with Confidence Intervals

Interpretation

- Forecast converges to $\bar{\sigma}$
- Uncertainty increases with forecast horizon



Rolling Forecast: Step-by-Step Prediction

Procedure

S&P 500, $W=500$, GARCH(1,1)-t

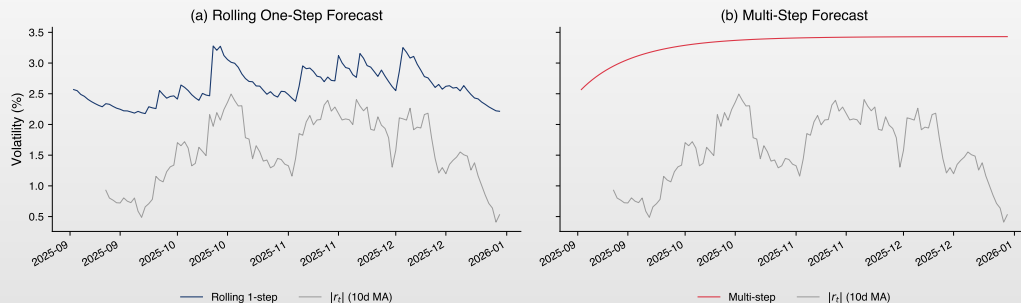
- Re-estimate GARCH on $[t-W, t-1]$; forecast $\hat{\sigma}_{t|t-1}$
- Compare with realized vol. (20-day rolling std.)

Results (2015 days OOS)

- $\rho = 0.938 \Rightarrow$ excellent tracking; MAE = 0.15%, RMSE = 0.24%
- COVID-19: temporary under-prediction, rapid adaptation

Rolling Forecast: Step-by-Step Prediction

Rolling vs Multi-Step GARCH Forecasts



 TSA_ch5_rolling_forecast

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Case Study: Bitcoin

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VaR Backtesting: Kupiec Test

Unconditional Coverage Test

Tests whether the observed violation rate equals the expected rate p (e.g., 1% for VaR 1%).

Let N = number of VaR violations, T = total observations, $\hat{p} = N/T$.

Likelihood Ratio Statistic:

$$LR_{uc} = -2 \ln \left[\frac{(1-p)^{T-N} p^N}{(1-\hat{p})^{T-N} \hat{p}^N} \right] \sim \chi^2(1)$$

Hypotheses

- $H_0: \hat{p} = p$ (VaR model is correctly calibrated)
- $H_1: \hat{p} \neq p$ (VaR model under- or over-estimates risk)

VaR Backtesting: Christoffersen Test

Conditional Coverage Test

Tests both **unconditional coverage** and **independence** of violations.
Violations should be independent — no clustering of exceptions!

Test Components

- ▣ **Independence test** (LR_{ind}): Tests if violations are serially independent
- ▣ **Conditional coverage**: $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$

Interpretation

- ▣ Reject LR_{uc} : wrong frequency
- ▣ Reject LR_{ind} : clustered violations
- ▣ Reject LR_{cc} : model fails

Full Backtesting: Results and Decision

Application S&P 500 ($T=500$, VaR 1%)

- ▣ Rolling window: $W = 500$ days, EGARCH(1,1)-t model
- ▣ Test window: $T - W$ days out-of-sample
- ▣ Compare forecasted VaR with realized return each day

Typical Output

- ▣ Number of violations vs expected $\Rightarrow$ Basel zone (green/yellow/red)
- ▣ Kupiec (frequency): $H_0: \hat{p} = \alpha \Rightarrow$ p-value $> 0.05 \Rightarrow$ accepted
- ▣ Christoffersen (independence): H_0 : violations i.i.d. $\Rightarrow$ p-value $> 0.05 \Rightarrow$ accepted
- ▣ Combined test: both conditions met $\Rightarrow$ **model validated**

Expected Shortfall Backtesting

Why Backtest ES?

- VaR only measures the *threshold* of extreme losses, not their *magnitude*
- ES = average loss *given* VaR is breached $\Rightarrow$ more informative for tail risk

Definition 24 (Acerbi-Székely Z-test (2014))

$Z_{\text{ES}} = \frac{1}{n\alpha} \sum_{t: r_t < -\text{VaR}_t} \frac{r_t}{\text{ES}_t} + 1$. Under H_0 (correct ES model): $\mathbb{E}[Z_{\text{ES}}] = 0$. Reject if $Z_{\text{ES}} < z_{\alpha/2}$.

Key Insight

- Unlike VaR backtesting (binary hit/miss), ES backtesting uses the *magnitude* of losses
- More powerful but requires larger samples ($T > 1000$); alt.: McNeil & Frey (2000) bootstrap
- ES under Student- t : $\text{ES}_{\alpha}^{t_{\nu}} = -\mu + \sigma \cdot \frac{\nu + (t_{\alpha, \nu})^2}{\nu - 1} \cdot \frac{f_{t_{\nu}}(t_{\alpha, \nu})}{\alpha}$ — heavier tails $\Rightarrow$ higher ES

Basel IV: From VaR to ES

Regulatory Shift

- ▣ **Basel II/III**: Capital based on VaR 99% (10-day)
- ▣ **Basel IV** (FRTB, 2023+): Capital based on **ES 97.5%** (10-day)
- ▣ Rationale: ES is **coherent** (subadditive), VaR is not

Equivalence under Normality

$$ES_{97.5\%}^{\text{Normal}} = \frac{\phi(z_{0.025})}{0.025} \cdot \sigma \approx 2.338 \sigma \approx VaR_{99\%}^{\text{Normal}} = 2.326 \sigma$$

⇒ Under normality, ES 97.5% $\approx$ VaR 99%. Under fat tails, ES $\gg$ VaR.

Practical Implication

- ▣ Student-*t* GARCH models become **essential** for accurate ES estimation
- ▣ Banks must now model the *entire tail*, not just a single quantile

ARMA-GARCH: Joint Mean and Variance Modeling

Why Joint Modeling?

Serial correlation $\Rightarrow$ ARMA for mean; **Volatility clustering** $\Rightarrow$ GARCH for variance.

Definition 25 (ARMA(p,q)-GARCH(r,s))

Mean equation: $r_t = \mu + \sum_{i=1}^p \phi_i(r_{t-i} - \mu) + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$

Variance equation: $\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \sum_{i=1}^r \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2$

ARMA-GARCH: Model Selection Strategy

Step-by-Step Approach

1. **Identify mean model:** Check ACF/PACF of returns for ARMA structure
2. **Test for ARCH effects:** Apply ARCH-LM test to residuals
3. **Specify variance model:** Usually GARCH(1,1) is sufficient
4. **Joint estimation:** Estimate both equations via MLE
5. **Diagnostic checking:** Standardized residuals should be i.i.d.

Common Specifications

- ▣ **Stock returns:** AR(1)-GARCH(1,1) or ARMA(1,1)-GARCH(1,1)
- ▣ **Exchange rates:** Often just GARCH(1,1) (no mean dynamics)
- ▣ **Interest rates:** AR(1)-EGARCH(1,1) for leverage effects

Python: GARCH Estimation with arch

Key Steps

```
from arch import arch_model
import numpy as np
from scipy.stats import norm
# Fit GARCH(1,1) with Student-t innovations
model = arch_model(returns, vol='Garch', p=1, q=1,
                   dist='StudentsT', mean='AR', lags=1)
result = model.fit(dispatch='off')
print(result.summary())
# Forecast 20-step ahead volatility
forecast = result.forecast(horizon=20)
sigma_forecast = np.sqrt(forecast.variance.iloc[-1])
# Compute 1-day VaR (1%) from conditional distribution
sigma_1 = sigma_forecast.iloc[0]
VaR_1pct = norm.ppf(0.01) * sigma_1 # Normal approx
```

Note: For Student-*t* VaR, replace `norm.ppf` with `scipy.stats.t.ppf(0.01, df=nu)` scaled by $\sqrt{(\nu - 2)/\nu}$.

Practical Checklist: GARCH Modeling (I)

Recommended Workflow — Steps 1–5

1. **Data:** Compute log-returns: $r_t = \ln(P_t/P_{t-1}) \times 100$
2. **Explore:** Check stylized facts — kurtosis $\gg 3$, significant $\text{ACF}(r_t^2)$, QQ-plot
3. **ARCH test:** Engle LM test on $\text{AR}(p)$ residuals — H_0 : no ARCH effects
4. **Estimate:** Start with GARCH(1,1)-t; compare with EGARCH, GJR-GARCH
5. **Distribution:** **Not** Normal — use **Student-t** (fat tails) or Skewed-t

Note

- ▣ Steps 1–3 are **mandatory** before any GARCH estimation
- ▣ Step 4: GARCH(1,1) is sufficient in 90% of cases — higher orders rarely improve forecasts

Practical Checklist: GARCH Modeling (II)

Recommended Workflow — Steps 6–9

6. **Persistence:** Check $\alpha + \beta$ — if ≈ 1 , consider IGARCH
7. **Diagnostics:** Standardized residuals $\hat{z}_t = r_t / \hat{\sigma}_t$ must be i.i.d.
8. **Selection:** AIC/BIC + Diebold-Mariano on out-of-sample forecasting
9. **Apply:** VaR/ES with rolling window (250–500 days) + Kupiec/Christoffersen backtesting

Common Pitfalls

- ▣ **Convergence:** If optimizer fails, switch algorithm (`method='Nelder-Mead'`) or starting values
- ▣ **Structural breaks:** On long series (2000–2024), GARCH parameters are **not constant** — rolling windows are essential

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Step 1: Data — S&P 500 Daily Returns

Data Description

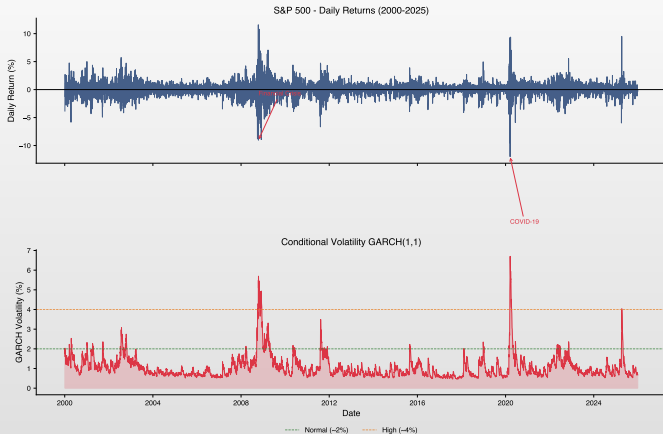
- ▣ **Source:** Yahoo Finance, S&P 500, daily data 2000–2024 ($T > 6000$)
- ▣ **Returns:** $r_t = \ln(P_t/P_{t-1}) \times 100$

Descriptive Statistics

Mean	Std. Dev.	Skewness	Kurtosis	Min	Max
0.034%	1.21%	−0.29	13.8	−12.8%	+11.0%

- ▣ Fat tails (kurtosis $\gg 3$) and negative skewness $\Rightarrow$ ARCH effects

Step 1: Data — S&P 500 Daily Returns



Step 2: Testing for ARCH Effects

Python Code — ARCH-LM and Ljung-Box on r_t^2

Results

Test	Statistic	p-value
ARCH-LM (10 lags)	892.4	< 0.0001
Ljung-Box r_t^2 (lag 20)	4217.6	< 0.0001

□ **Conclusion:** Strong ARCH effects $\Rightarrow$ significant heteroskedasticity

Step 3: Estimated Parameters — Comparison

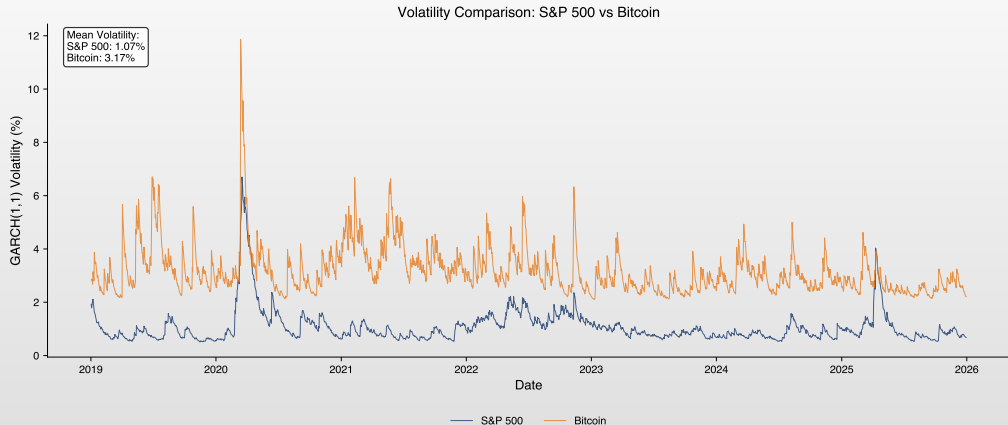
Estimated Parameters Table

Model	ω	α	β	γ	$\alpha + \beta$	ν	HL
GARCH-N	0.011	0.088	0.901	—	0.989	—	60 days
GARCH-t	0.011	0.088	0.900	—	0.989	6.42	60 days
EGARCH-t	0.003	0.103	0.987	-0.120	—	6.38	53 days
GJR-t	0.010	0.022	0.906	0.126	0.991	6.51	78 days

Interpretation

- EGARCH $\gamma = -0.12$ **significant** $\Rightarrow$ **leverage effect confirmed**
- GJR: $\alpha_{\text{neg}} = \alpha + \gamma = 0.148$ vs $\alpha_{\text{pos}} = 0.022 \Rightarrow$ **strong asymmetry**

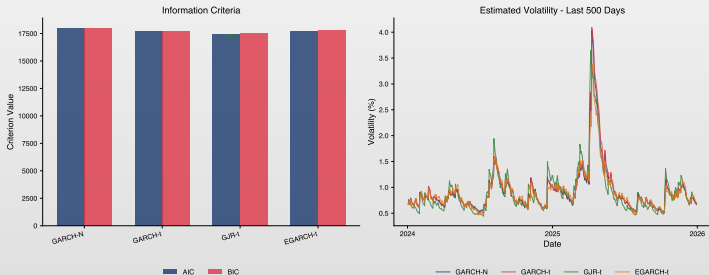
Step 3: Estimated Conditional Volatility — S&P 500



Step 4: Leverage Effect — Visualization

GARCH vs EGARCH — Volatility Differences

- EGARCH produces **higher** volatility after negative shocks (2008, 2020)
- Symmetric GARCH **underestimates** risk during crisis periods
- Difference: up to 2–3 percentage points in daily volatility



Step 5: Model Selection — AIC/BIC

Information Criteria

Model	Log-Lik	AIC	BIC	Rank
GARCH(1,1)-N	-8042.3	16090.6	16111.0	4
GARCH(1,1)-t	-7981.5	15971.0	15997.8	3
EGARCH(1,1)-t	-7964.2	15938.4	15971.6	1
GJR-GARCH(1,1)-t	-7968.1	15946.2	15979.4	2

Decision

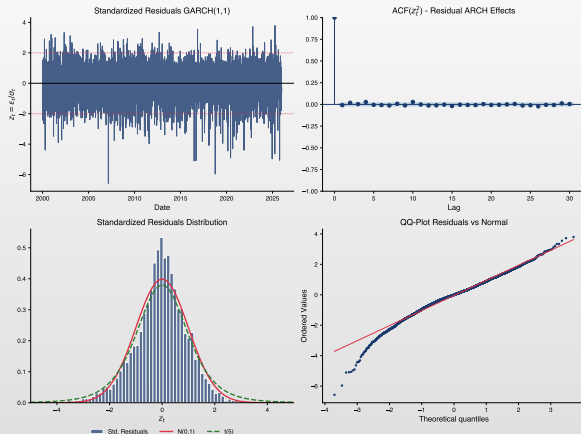
- **EGARCH(1,1)-t wins:** lowest AIC and BIC
- Student-t superior to Normal ($\Delta\text{AIC} \approx 120$) $\Rightarrow$ fat tails matter!
- Leverage effect justifies asymmetric models ($\Delta\text{AIC} \approx 33$ vs GARCH-t)

Step 6: Diagnostics — EGARCH(1,1)-t

Checks on Standardized Residuals $z_t = \varepsilon_t / \hat{\sigma}_t$

- ▣ **Ljung-Box** on z_t : p-value = 0.38 — no residual autocorrelation
- ▣ **Ljung-Box** on z_t^2 : p-value = 0.52 — **ARCH effects eliminated**
- ▣ **Q-Q plot**: points follow the theoretical Student-t line
- ▣ **Conclusion**: EGARCH(1,1)-t adequately captures volatility dynamics

Step 6: Diagnostics — EGARCH(1,1)-t

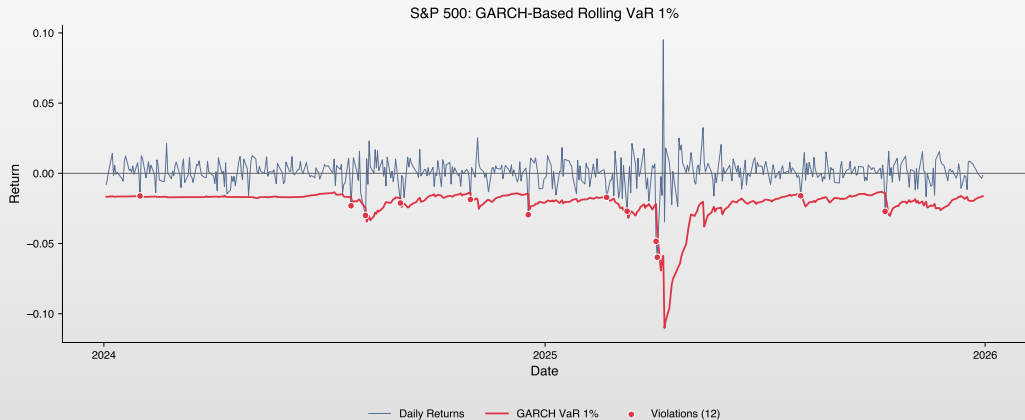


Step 7: Backtesting Rolling VaR — S&P 500

Kupiec + Christoffersen Results (2015 days out-of-sample)

Test	Statistic	p-value	Decision
Violations	27/2015 ($\hat{p} = 1.34\%$)	—	Green zone
Kupiec (uc)	2.13	0.145	Accepted
Christoffersen (ind)	0.79	0.375	Accepted
Combined (cc)	2.91	0.233	Accepted

Step 7: Rolling VaR — S&P 500



Step 8: Conclusions — S&P 500 Case Study

Step-by-Step Methodology Summary

1. **Data:** log returns, descriptive statistics $\Rightarrow$ fat tails, skewness
2. **ARCH test:** ARCH-LM + Ljung-Box on $r_t^2 \Rightarrow$ significant ARCH effects
3. **Estimation:** 4 candidate models (symmetric/asymmetric $\times$ Normal/Student-t)
4. **Selection:** AIC/BIC $\Rightarrow$ **EGARCH(1,1)-t** winner
5. **Diagnostics:** standardized residuals $\Rightarrow$ model adequate
6. **VaR:** rolling window + Kupiec/Christoffersen backtesting $\Rightarrow$ model **validated**

Key Lessons

- ▣ Student-t distribution is **essential** for financial data
- ▣ Leverage effect: asymmetric models **mandatory** for equities
- ▣ Systematic backtesting: not just “looks good”, but **statistically tested**
- ▣ **Even though returns are hard to predict, their volatility is often predictable**

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Step 1: Data — Bitcoin Daily Returns

Data Description

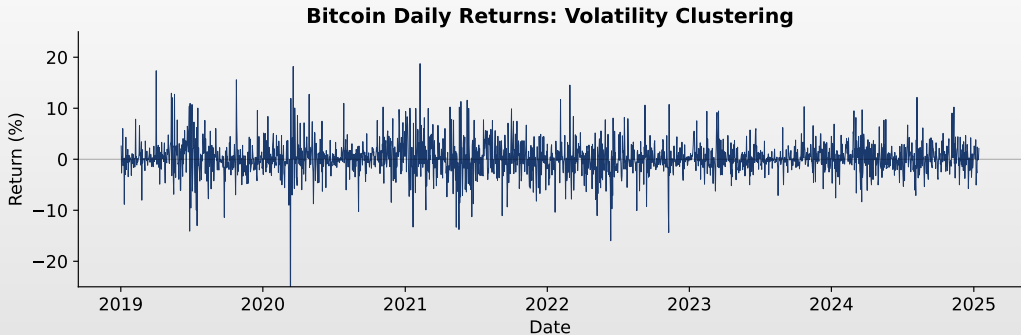
- ▣ **Source:** Yahoo Finance (BTC-USD), daily data 2018–2024
- ▣ Log returns: mean $\approx 0.05\%$, volatility $\approx 3.5\%$

Descriptive Statistics

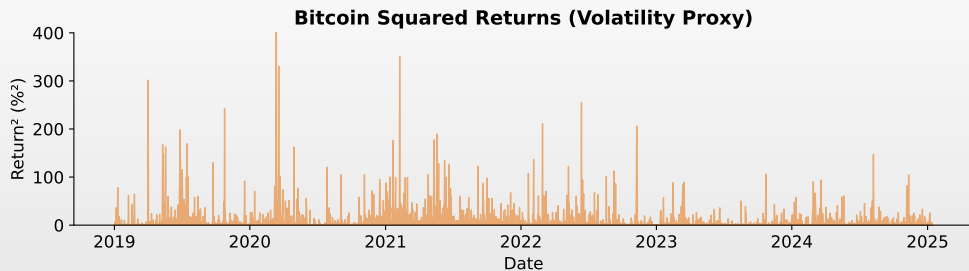
Mean	Std. Dev.	Skewness	Kurtosis	Min	Max
0.05%	3.48%	-0.72	12.1	-46.5%	+22.5%

- ▣ Volatility $\sim 3\times$ higher than S&P 500
- ▣ Extreme kurtosis — high risk of large losses

Step 1: Data — Bitcoin Daily Returns



Step 2: Testing for ARCH Effects — Bitcoin

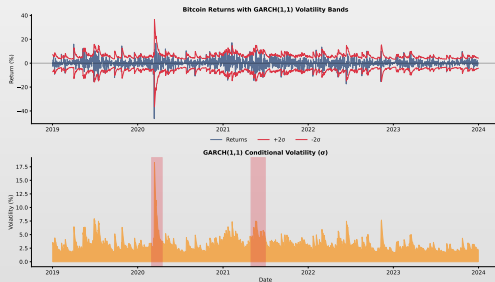


 TSA_ch5_btc_arch

Step 3: Conditional Volatility — Bitcoin

GJR-GARCH(1,1)-t Diagnostics

- ▣ Ljung-Box on z_t^2 : p-value = 0.41 — **ARCH effects eliminated**
- ▣ Volatility peaks: March 2020 (COVID), May 2022 (Terra/Luna)
- ▣ Daily volatility: from 1% (calm periods) to >15% (crises)



Step 4: Estimation and Model Selection — Bitcoin

Estimated Parameters

Model	ω	α	β	γ	$\alpha + \beta$	ν	AIC
GARCH-t	0.42	0.131	0.848	—	0.979	4.82	9284
EGARCH-t	0.08	0.184	0.976	-0.061	—	4.79	9276
GJR-t	0.40	0.088	0.854	0.078	0.976	4.85	9271

Interpretation

- **GJR-GARCH-t wins** (lowest AIC)
- $\nu \approx 4.8$: **much heavier tails** than S&P 500 ($\nu = 6.4$)
- $\alpha = 0.131$ (BTC) vs 0.088 (S&P) — Bitcoin reacts faster to news
- Leverage effect weaker than for stocks ($\gamma_{\text{BTC}} = 0.078$ vs 0.126)

Step 5: Backtesting Rolling VaR — Bitcoin

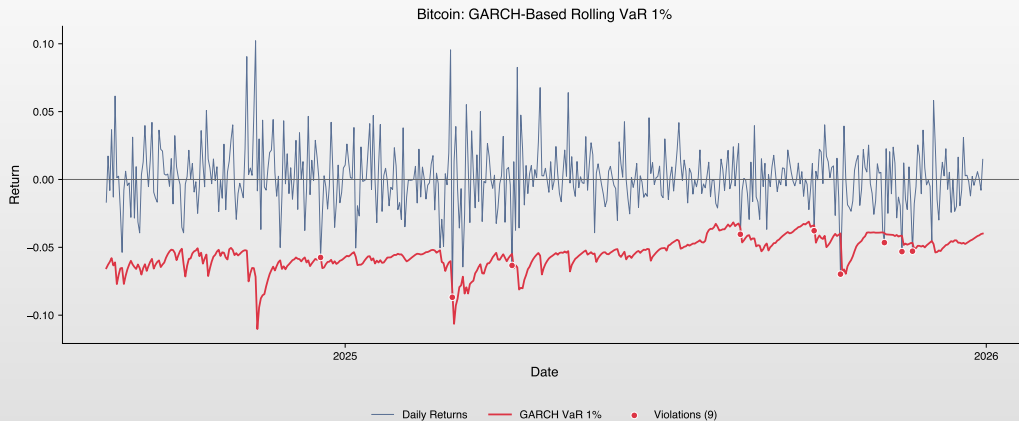
Statistical Tests (2421 days out-of-sample)

Test	Statistic	p-value	Decision
Violations	28/2421 ($\hat{p} = 1.16\%$)	—	Green zone
Kupiec (uc)	0.57	0.450	Accepted
Christoffersen (ind)	0.94	0.333	Accepted
Combined (cc)	1.51	0.471	Accepted

Interpretation

- Volatility ranges from 3% to 38% — rolling window is **essential**
- All tests **accepted**: model valid for risk management

Step 5: Rolling VaR — Bitcoin



Step 5: VaR Rolling Window — Bitcoin

Rolling Window GJR-GARCH-t (W=500 days, VaR 1%)

Rolling VaR Characteristics Bitcoin (2018–2024)

- Mean VaR: 9.34% ($\approx$ EUR 93,400 / 1M EUR)
- Max VaR: 37.54% $\Rightarrow$ COVID crash March 2020
- Min VaR: 2.90% $\Rightarrow$ calm period
- Bitcoin: rolling VaR $\sim 4\times$ larger than S&P 500 at same exposure

Final Comparison: S&P 500 vs Bitcoin

Comparative Summary

	S&P 500	Bitcoin
Average volatility	1.2%	3.5%
Kurtosis	13.8	12.1
Student-t ν	6.42	4.82
Best model	EGARCH(1,1)-t	GJR-GARCH(1,1)-t
Leverage effect	Strong ($\gamma = -0.12$)	Moderate ($\gamma = 0.078$)
Half-life	~60 days	~42 days
Rolling VaR 1% mean	2.53%	9.34%
Rolling VaR 1% max	22.02% (COVID)	37.54% (COVID)
Rolling ES 1% mean	3.42%	12.87%
Kupiec	Accepted (p=0.145)	Accepted (p=0.450)
Christoffersen (ind)	Accepted (p=0.375)	Accepted (p=0.333)

General Conclusion

- Re-estimating GARCH at each step: Kupiec + Christoffersen **accepted**
- Rolling window VaR: **mandatory** — static VaR is completely inadequate
- Student-t + asymmetric model: **essential** for both markets

Final Comparison: S&P 500 vs Bitcoin (cont.)

General Conclusion

- ▣ Re-estimating GARCH at each step: Kupiec + Christoffersen **accepted**
- ▣ Rolling window VaR: **mandatory** $\Rightarrow$ static VaR is completely inadequate
- ▣ Student-t + asymmetric model: **essential** for both markets

Key Formulas

Volatility Models

- ▣ **ARCH(q):** $\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$
- ▣ **GARCH(1,1):** $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
- ▣ **EGARCH:** $\ln(\sigma_t^2) = \omega + \alpha(|z_{t-1}| - \mathbb{E}[|z|]) + \gamma z_{t-1} + \beta \ln(\sigma_{t-1}^2)$
- ▣ **GJR-GARCH:** $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I_{t-1} + \beta \sigma_{t-1}^2$

Properties and Measures

- ▣ **Unconditional variance:** $\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$ **Half-life:** $HL = \frac{\ln(0.5)}{\ln(\alpha+\beta)}$
- ▣ **VaR:** $\text{VaR}_\alpha = z_\alpha \cdot \sigma_{T+1}$ **Stationarity:** $\alpha + \beta < 1$
- ▣ **ARCH-LM:** $LM = T \cdot R^2 \sim \chi^2(q)$

AI Exercise: Critical Thinking

Prompt to test in ChatGPT / Claude / Copilot

"Using yfinance, download daily closing prices for S&P 500 (^GSPC) from 2018-01-01 to 2024-12-31 (approx. 1,750 observations). Compute daily log returns. Test for ARCH effects, fit a GARCH(1,1) model, and forecast volatility for the next 20 trading days. Give me complete Python code."

Exercise:

1. Run the prompt in an LLM of your choice and critically analyze the response
2. Does it compute log returns correctly? Does it remove the mean before fitting GARCH?
3. How does it test for ARCH effects? Does it use Engle's LM test?
4. Does it separate the mean equation from the variance equation?
5. Does it discuss asymmetric effects (GJR-GARCH, EGARCH)?

AI Exercise: Critical Thinking (cont.)

Exercise (continued):

6. Are VaR estimates computed from the conditional distribution?
7. Identify at least **two factual errors** or questionable assumptions in the LLM output
8. Does the code verify the stationarity constraint $\alpha + \beta < 1$? If not, add it
9. Compare the LLM's VaR estimate with the results from this chapter. Are they consistent?

Warning: AI-generated code may run without errors and look professional. *That does not mean it is correct.*

Summary – Chapter 5: Volatility Models

Key Concepts

- ▣ **ARCH(q)**: conditional variance depends on past squared errors (Nobel 2003)
- ▣ **GARCH(p,q)**: adds variance lags for persistence (GARCH(1,1) in 90% of cases)
- ▣ **EGARCH**: allows leverage effect, no positivity constraints
- ▣ **GJR-GARCH/TGARCH**: captures asymmetry with indicator variables

Applications

- ▣ Risk measurement and forecasting (VaR, ES)
- ▣ Derivative pricing, dynamic hedging, portfolio management

Practical Tip

- ▣ Start with GARCH(1,1), check for leverage, choose distribution minimizing AIC/BIC!
 - ▶ Student-t often superior to normal distribution

What's Next?

Chapter 5b: Multivariate GARCH Models

- ▣ **Conditional covariance:** modeling the joint dynamics of multiple assets
- ▣ **DCC-GARCH** (Dynamic Conditional Correlation — Engle, 2002): dynamic conditional correlations
- ▣ **BEKK** (Baba-Engle-Kraft-Kroner, 1995): guaranteeing positivity of H_t
- ▣ **CCC-GARCH** (Constant Conditional Correlation — Bollerslev, 1990): constant conditional correlations
- ▣ **Applications:** optimal hedging, portfolio allocation, financial contagion

Questions?

Question 1

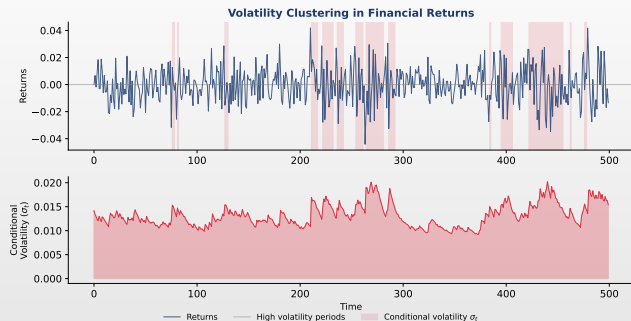
Question

- ☐ What best describes the phenomenon of *volatility clustering* in financial series?

Answer Choices

- (A) Financial returns are normally distributed and independent
- (B) Periods of high volatility are followed by high volatility, and vice versa
- (C) Volatility is constant over time (homoskedasticity)
- (D) Correlation between returns is always positive

Question 1: Answer



Answer: (B)

- Volatility clustering is a fundamental stylized fact of financial series. It implies that conditional variance is **predictable**, motivating ARCH/GARCH models.

Question 2

Question

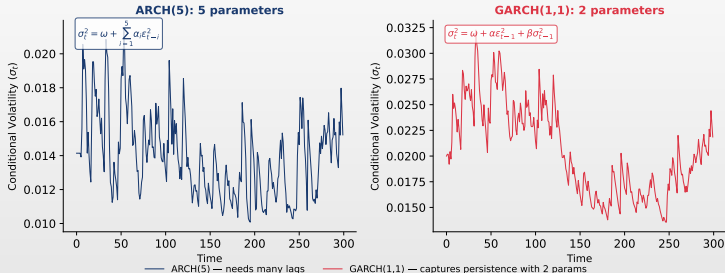
□ What is the main difference between an ARCH(q) and a GARCH(p,q) model?

Answer Choices

- (A) GARCH models the conditional mean, ARCH models the variance
- (B) ARCH includes lags of conditional variance, GARCH does not
- (C) GARCH adds lags of conditional variance (σ_{t-j}^2) beyond squared errors
- (D) ARCH is more parsimonious than GARCH

Question 2: Answer

ARCH vs GARCH: Parsimony



Answer: (C)

- GARCH(1,1) captures the same persistence as ARCH(q) with only 2 parameters instead of q . In practice, GARCH(1,1) is sufficient in 90% of cases.

Question 3

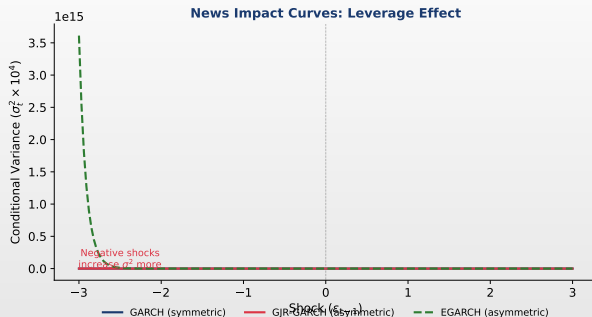
Question

- ☐ What is the *leverage effect* and which GARCH models capture it?

Answer Choices

- (A) Positive shocks increase volatility more; captured by standard GARCH
- (B) Negative shocks increase volatility more; captured by EGARCH and GJR-GARCH
- (C) Volatility is symmetric; captured by all GARCH models
- (D) Financial leverage effect on stock prices; captured by IGARCH

Question 3: Answer



Answer: (B)

- Price drops increase volatility **more** than equivalent price increases. Standard GARCH uses ε_{t-1}^2 , losing the sign information.

Question 4

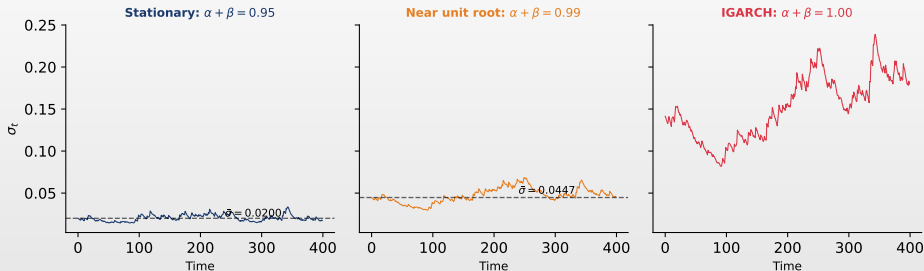
Question

□ What is the stationarity condition for a GARCH(1,1) model?

Answer Choices

- (A) $\alpha + \beta = 1$
- (B) $\alpha > 0$ and $\beta > 0$
- (C) $\alpha + \beta < 1$, with $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$
- (D) $\alpha \cdot \beta < 1$

Question 4: Answer

GARCH(1,1) Stationarity: $\alpha + \beta < 1$ 

Answer: (C)

- This ensures a finite unconditional variance $\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$. When $\alpha + \beta = 1$ (IGARCH), variance is infinite.

Question 5

Applied Computational Question

A GARCH(1,1) model estimated on daily log returns (%) gives: $\omega = 0.05$, $\alpha = 0.15$, $\beta = 0.82$.

- (A) Compute the unconditional variance $\bar{\sigma}^2$ and annualized volatility.
 - (B) Compute the half-life of a volatility shock.
 - (C) For a EUR 2 000 000 portfolio, compute the 1-day VaR at 1% under:
 - ☐ Normal distribution
 - ☐ Student- t with $\nu = 5$ degrees of freedom
- (Use $\sigma_{T+1} = \bar{\sigma}$ for simplicity.)

Question 5: Answer

Solution

(A) $\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta} = \frac{0.05}{1-0.97} = 1.667 \Rightarrow \bar{\sigma} = 1.291\% \text{ daily} \quad \sigma_{\text{ann}} = 1.291\% \times \sqrt{252} \approx 20.5\%$

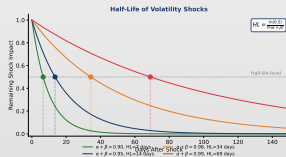
(B) $HL = \frac{\ln 0.5}{\ln(\alpha+\beta)} = \frac{-0.693}{\ln(0.97)} = \frac{-0.693}{-0.0305} \approx 23 \text{ days}$

(C) Portfolio value $V = 2,000,000$ EUR, $\sigma_{T+1} = 1.291\%$:

Normal: $\text{VaR}_{1\%} = 2.326 \times 0.01291 \times 2,000,000 = 60,066$ EUR

Student-t(5): $t_{0.01,5} = 3.365$ scaled by $\sqrt{(\nu-2)/\nu} = \sqrt{3/5}$:

$\text{VaR}_{1\%} = \frac{3.365}{\sqrt{5/3}} \times 0.01291 \times 2,000,000 = 67,300$ EUR (+12% vs Normal!)



References I

Fundamental ARCH/GARCH Papers

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- Bollerslev, T. (1986). Generalized Autoregressive Conditional Heteroskedasticity, *Journal of Econometrics*, 31(3), 307–327.
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Asymmetric Models and Extensions

- Glosten, L.R., Jagannathan, R., & Runkle, D.E. (1993). On the Relation Between the Expected Value and the Volatility of the Nominal Excess Return on Stocks, *Journal of Finance*, 48(5), 1779–1801.
- Francq, C., & Zakoïan, J.-M. (2019). *GARCH Models: Structure, Statistical Inference and Financial Applications*, 2nd ed., Wiley.

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Textbooks and Financial Applications

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- ▣ Franke, J., Härdle, W.K., & Hafner, C.M. (2019). *Statistics of Financial Markets*, 4th ed., Springer.
- ▣ McNeil, A.J., Frey, R., & Embrechts, P. (2015). *Quantitative Risk Management*, 2nd ed., Princeton University Press.

Online Resources and Code

- ▣ **Quantlet**: <https://quantlet.com> – Code platform for quantitative methods
- ▣ **Quantinar**: <https://quantinar.com> – Learning platform for quantitative methods
- ▣ **GitHub TSA**: https://github.com/QuantLet/TSA/tree/main/TSA_ch5 – Python code for this chapter

References III

Estimation and Evaluation

- White, H. (1982). Maximum Likelihood Estimation of Misspecified Models, *Econometrica*, 50(1), 1–25.
- Hansen, P.R., & Lunde, A. (2005). A Forecast Comparison of Volatility Models: Does Anything Beat a GARCH(1,1)?, *Journal of Applied Econometrics*, 20(7), 873–889.
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High-Frequency and Risk Management

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References IV — Realized Volatility

High-Frequency and Realized Volatility

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Textbooks and Monographs

- ▣ Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed., Wiley.
- ▣ Francq, C., & Zakoïan, J.-M. (2019). *GARCH Models*, 2nd ed., Wiley.
- ▣ McNeil, A.J., Frey, R., & Embrechts, P. (2015). *Quantitative Risk Management*, 2nd ed., Princeton UP.
- ▣ Franke, J., Härdle, W.K., & Hafner, C.M. (2019). *Statistics of Financial Markets*, 4th ed., Springer.

References V — Additional Resources

Additional References

- ▣ Hamilton, J.D. (1994). *Time Series Analysis*, Princeton UP.
- ▣ Bollerslev, T., Engle, R.F., & Nelson, D.B. (1994). ARCH Models, in *Handbook of Econometrics*, Vol. 4, Elsevier.
- ▣ Patton, A.J. (2011). Volatility Forecast Comparison Using Imperfect Volatility Proxies, *J. Econometrics*, 160(1).

Online Resources and Code

- ▣ **Quantlet**: <https://quantlet.com> — Platform for quantitative methods code
- ▣ **Quantinar**: <https://quantinar.com> — Learning platform for quantitative methods
- ▣ **GitHub TSA**: https://github.com/QuantLet/TSA/tree/main/TSA_ch5 — Python code for this chapter
- ▣ **Online course**: <https://danpele.github.io/Time-Series-Analysis/> — Complete course materials

Competencies Acquired

By the end of the chapter, the student should be able to:

1. **Recognize** ARCH effects in data (ARCH-LM, Ljung-Box on r_t^2 , Q-Q plot)
2. **Estimate and interpret** a GARCH(1,1) — the role of α , β , unconditional variance, half-life
3. **Decide** whether asymmetry is needed (leverage effect $\Rightarrow$ EGARCH / GJR-GARCH)
4. **Choose** an innovation distribution (Normal vs Student-t vs GED)
5. **Produce** a volatility forecast (rolling window, convergence to $\bar{\sigma}^2$)
6. **Calculate and validate** a VaR (Kupiec + Christoffersen backtesting)

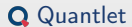
Central Message of the Chapter

- **Even though returns are hard to predict, their volatility is often predictable.**
- This is the fundamental idea that justifies the entire econometrics of volatility.

Thank You!

Questions?

Course materials available at: <https://danpele.github.io/Time-Series-Analysis/>



Quantlet



Quantinar